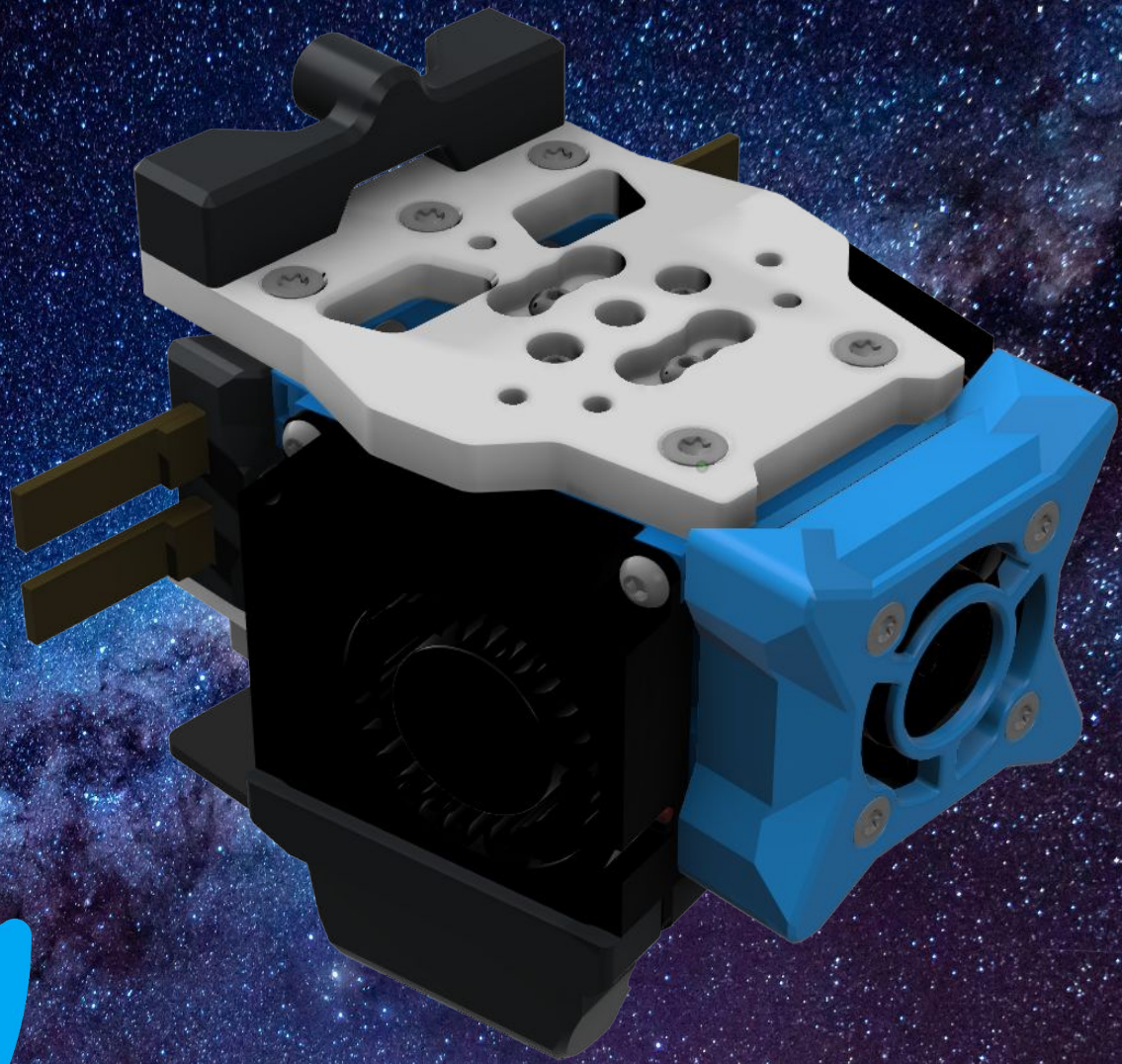


HYPERNOVA BUILD GUIDE

"We build precision with simplicity, so anyone can create the impossible."

FINAL VERSION • 2026-11-10

Before getting started | Kit checklist

Check every item before starting. Sort screws by size and head type.

SCREWS & STANDOFFS

☐	M3 × 40 countersunk Torx	**5**
☐	M3 × 10 countersunk Torx	**3**
☐	M3 × 20 button head Torx	**4**
☐	M3 × 12 countersunk Torx	**4**
☐	M3 × 8 button head Torx	**10**
☐	M3 Beacon standoff†	**2**
☐	M3 × 8 countersunk Torx	**6**
☐	M3 × 8 ultra low profile Torx	**2**
☐	M2.5 × 6 button head Torx	**8**
☐	M2 × 10 socket head	**2**
☐	M2.5 × 16 countersunk Torx	**4**

INSERTS

☐	M2.5 inserts (8 used + 2 spares)	**10**
☐	M3 heat-set inserts	**13**
☐	M3 HYPERSETS	**3**

FANS & PLATES

☐	4010 fans	**2**
☐	2510 Hyper Fan	**1**
☐	Top plate HN-TTPU	**1**
☐	Bottom plate HN-BPU	**1**
☐	MakerBeam XL T-nuts	**2**
☐	25 mm belt-tie spacers	**2**

†Beacon: 26 mm for UHF-length hotends; 30 mm for Chube.

Included extras: M3 × 10, M3 × 8, MakerBeam XL T-nuts and ultra low profile screws.

Prepare your tools and parts

TOOLS

Matching Torx and hex drivers

Temperature-controlled insert tool

Tweezers or pliers for handling inserts

Calipers and a small parts tray

PREPARE

Match your printed parts to your extruder and mount variant.

Inspect prints and clear the mounting holes.

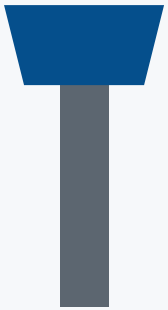
Compare the kit with the checklist before assembly.

Keep power disconnected during mechanical assembly.

HEAT-SET INSERTS: Set the installation tool to 250°C for ABS.

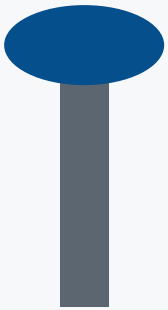
Identify the hardware before assembly

Compare the head shape as well as the thread size and length. Profiles below are schematic, not to scale.

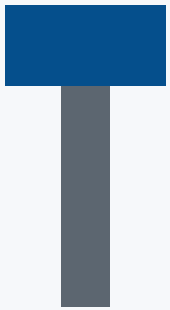


COUNTERSUNK

Ultra low profile screws have a shallower head. Keep them separate from the other M3 × 8 screws.



BUTTON HEAD



SOCKET HEAD

HYPERSETS are flanged heat-set inserts. Match the flange and insert type to the illustrated location.

Contents | Jump to a section

Print settings	02
Heat-set inserts	03
Core mounting	04
Top plate assembly	05
Hotend	06
Beacon and bottom plate	07
Fans	08
Belts	09
References	11

Preparation / hardware ID: preceding pages Final inspection: after Belts Installed

Print settings | HYPERNOVA parts

Use the settings below for the printed parts.

3D PRINTING PROCESS
Fused Deposition Modeling (FDM)

MATERIAL
ABS/ASA

LAYER HEIGHT
Recommended: 0.2mm

EXTRUSION WIDTH
Recommended: Forced 0.4mm

INFILL TYPE
Grid, Gyroid, Honeycomb, Triangle or Cubic

INFILL PERCENTAGE
Recommended: 50%

WALL COUNT
Recommended: 4

SOLID TOP/BOTTOM LAYERS
Recommended: 5

NovaCore | Side inserts

HARDWARE

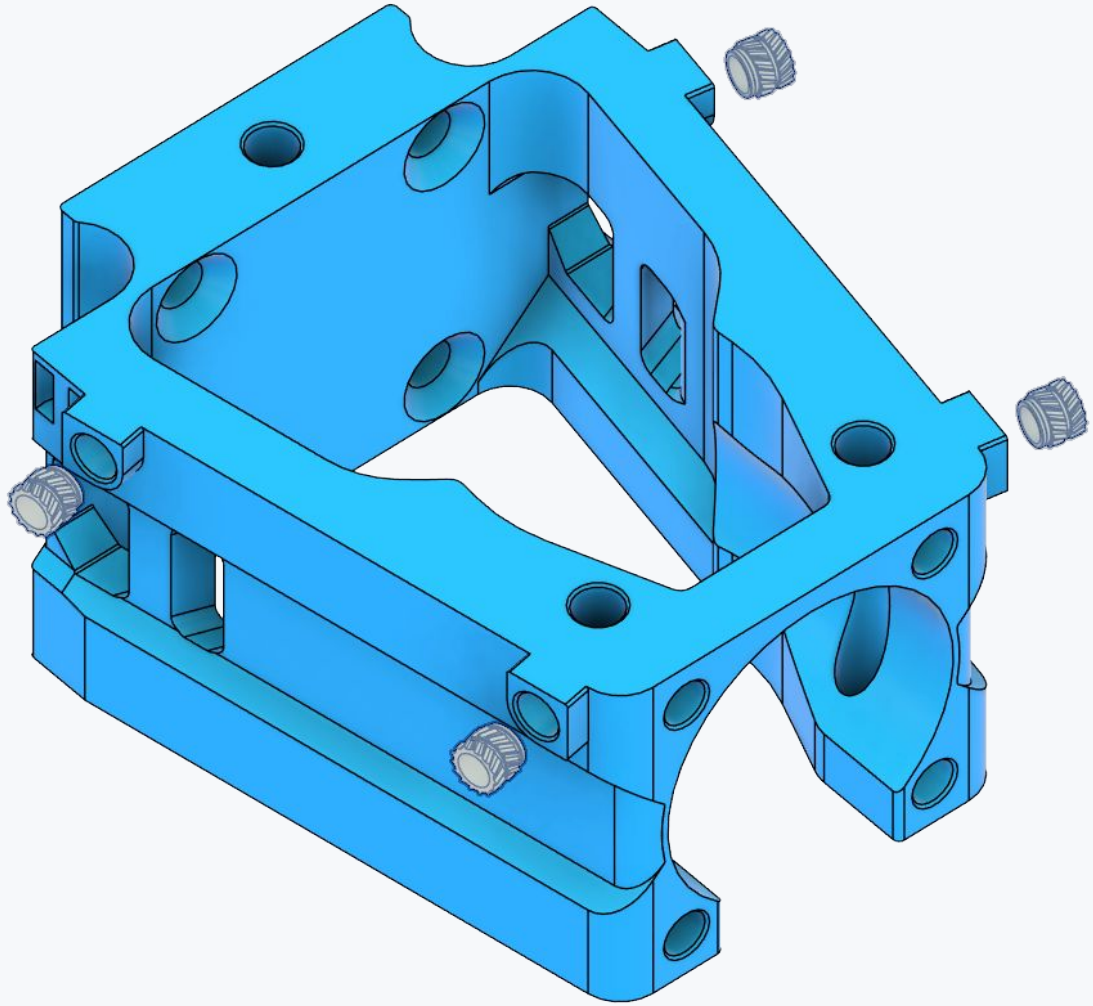
4 × M2.5 heat-set inserts

ACTION

Install the inserts in the side holes shown.

CHECK

Keep each insert square to the surface.



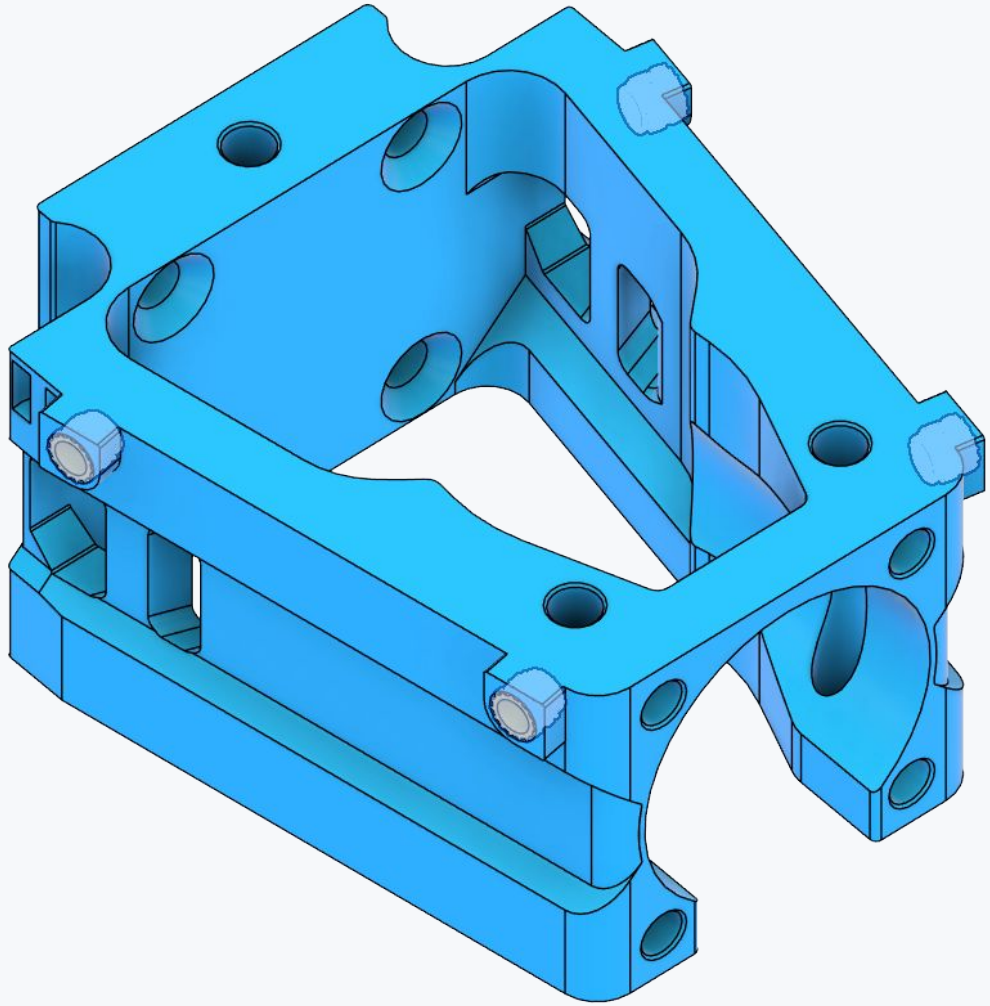
HEAT-SET INSERTS: Use a 250°C installation-tool setting for ABS.

NovaCore | Check the side inserts

CHECK

All four inserts sit flush and square.

Inspect from the side for tilted or protruding inserts.



NovaCore | Front inserts

HARDWARE

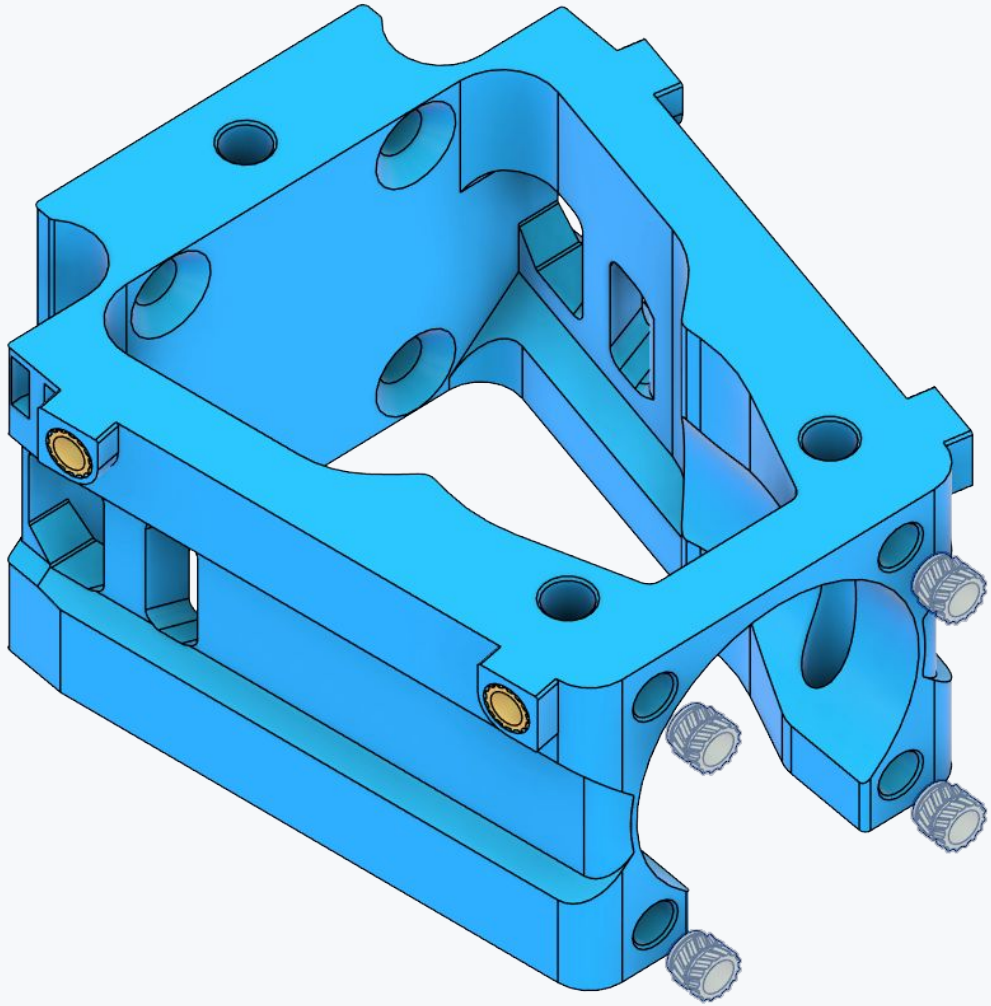
4 × M2.5 heat-set inserts

ACTION

Install the inserts in the front holes shown.

CHECK

Keep each insert square to the surface.



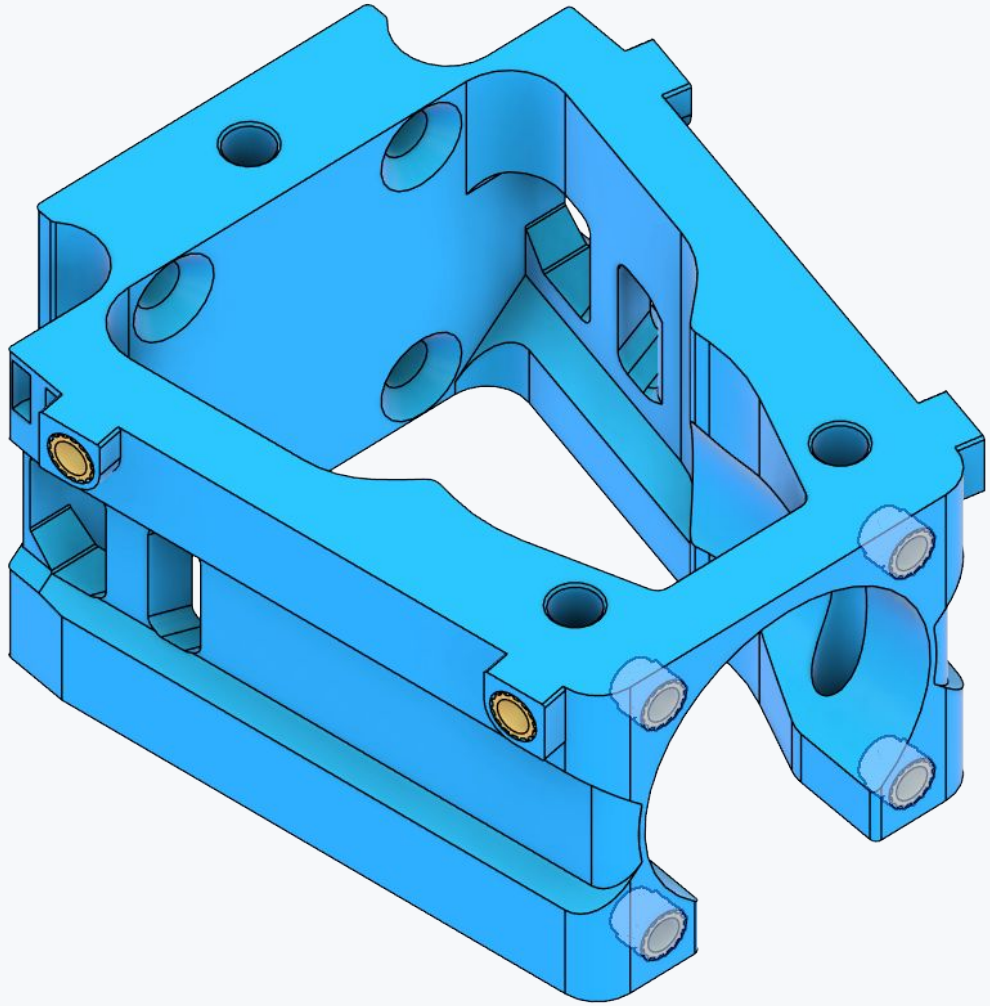
HEAT-SET INSERTS: Use a 250°C installation-tool setting for ABS.

NovaCore | Check the front inserts

CHECK

All four inserts sit flush and square.

Compare the installed positions with this view.



NovaCore | Install M3 heat-set inserts

HARDWARE

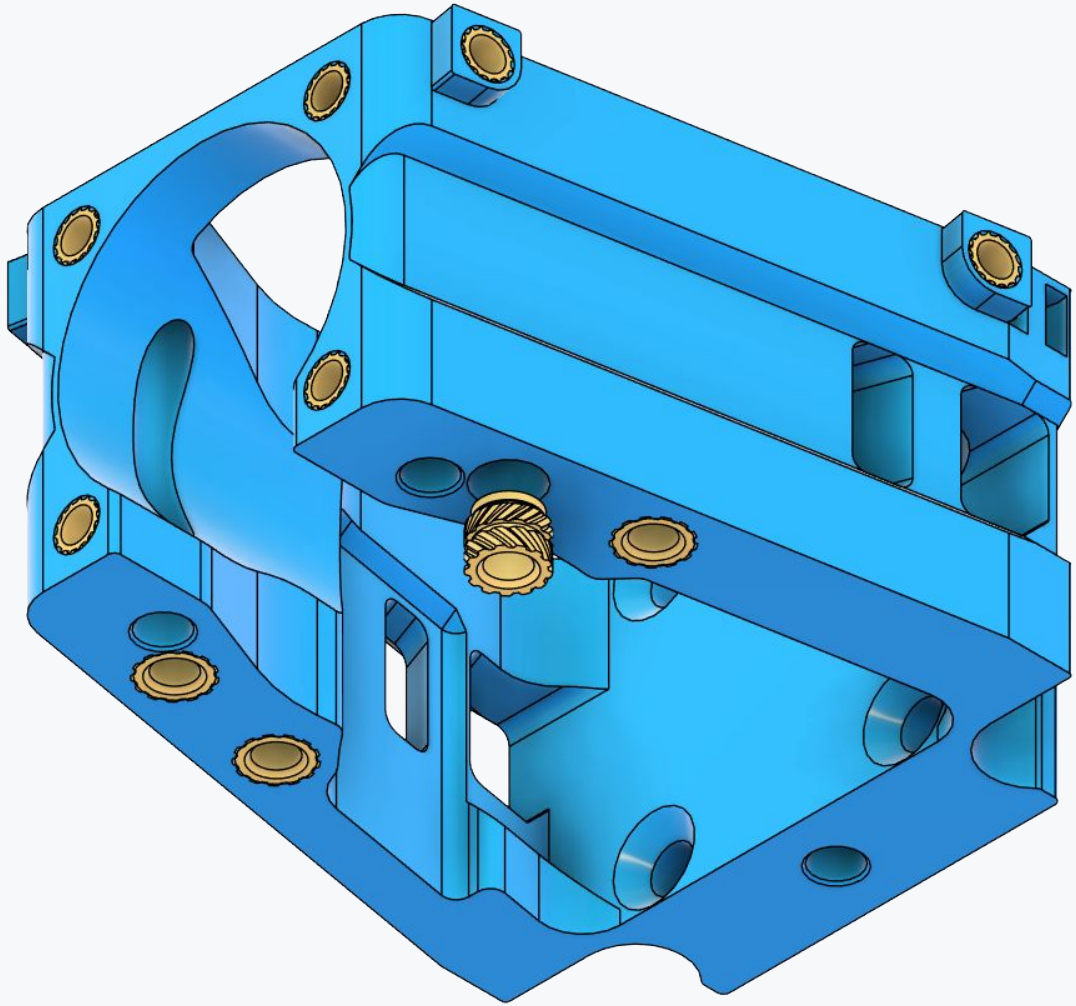
4 × M3 heat-set inserts

ACTION

Install the inserts in the four illustrated positions in the core.

CHECK

Keep each insert square and seated.



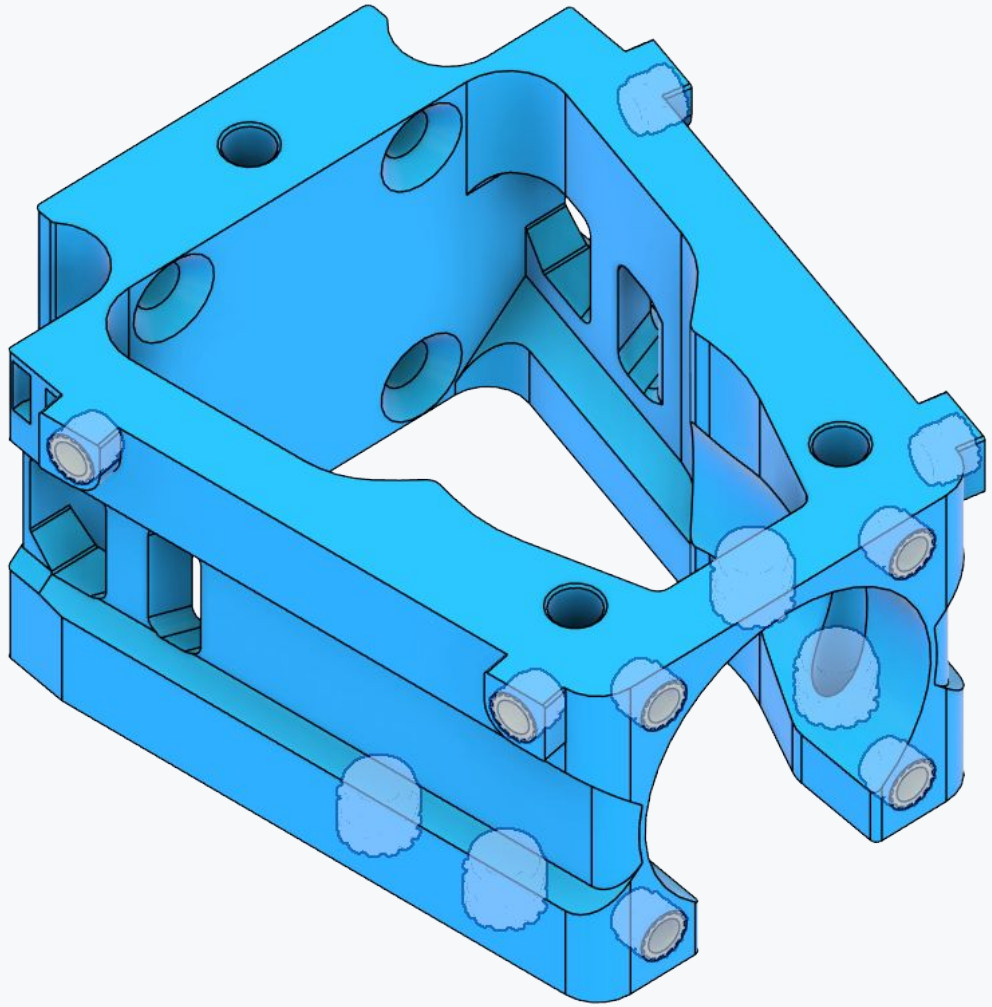
NovaCore | Completed insert check

CHECK

Compare the core with this completed view.

Check that the inserts are square and seated.

Confirm all four M3 inserts from step 3.5 are installed.



Spacer | Install HYPERSETS

HARDWARE

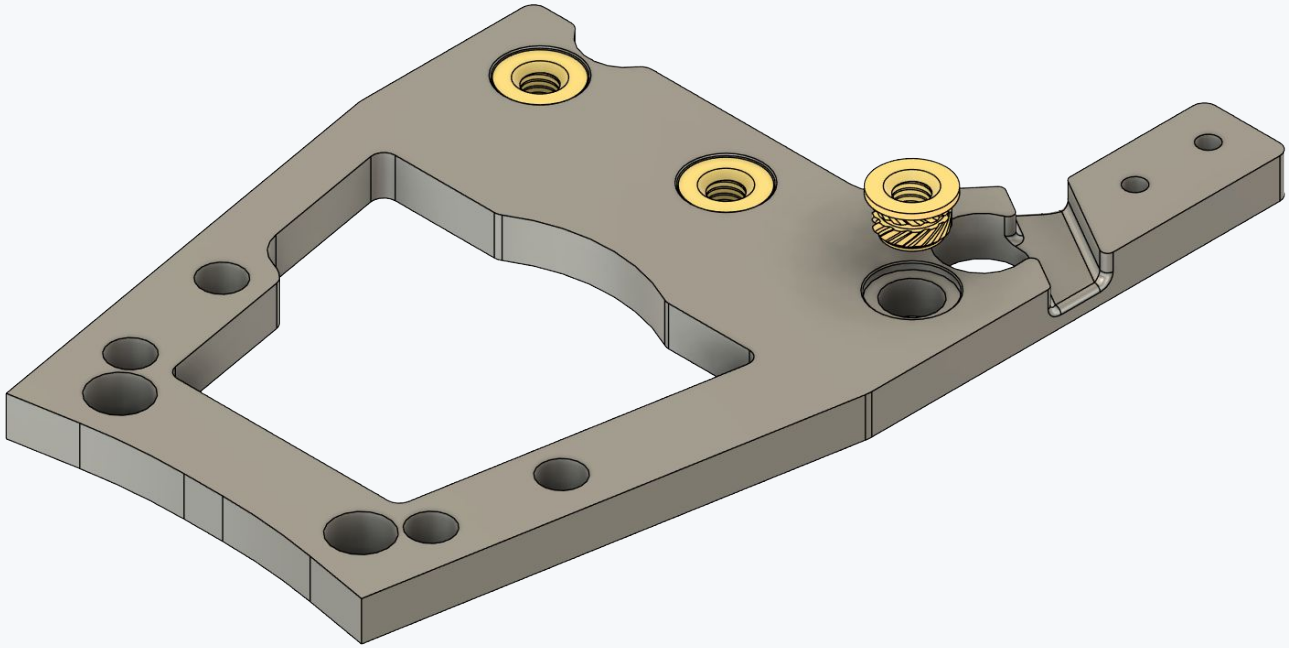
3 × M3 HYPERSETS

ACTION

Install them in the positions shown.

NOTE

HYPERSETS are flanged heat-set inserts.



Spacer | Install heat-set inserts

HARDWARE

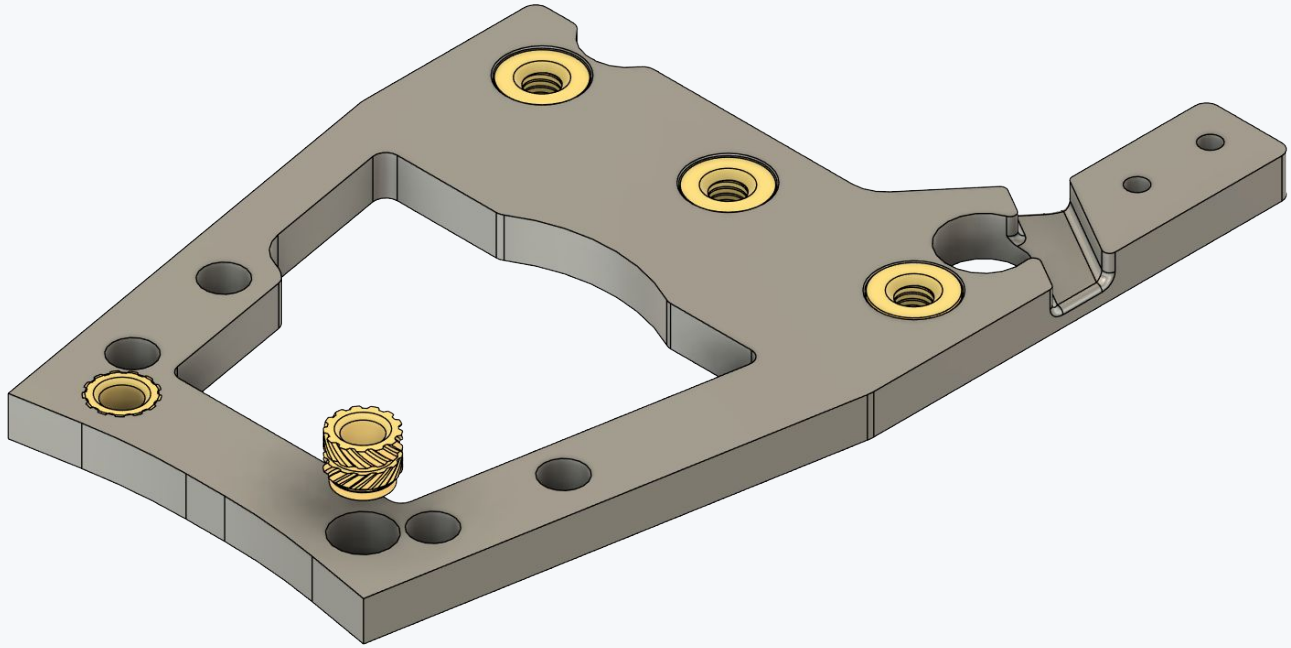
2 × M3 heat-set inserts

ACTION

Install them in the remaining illustrated positions.

CHECK

Match the completed view on the next page.

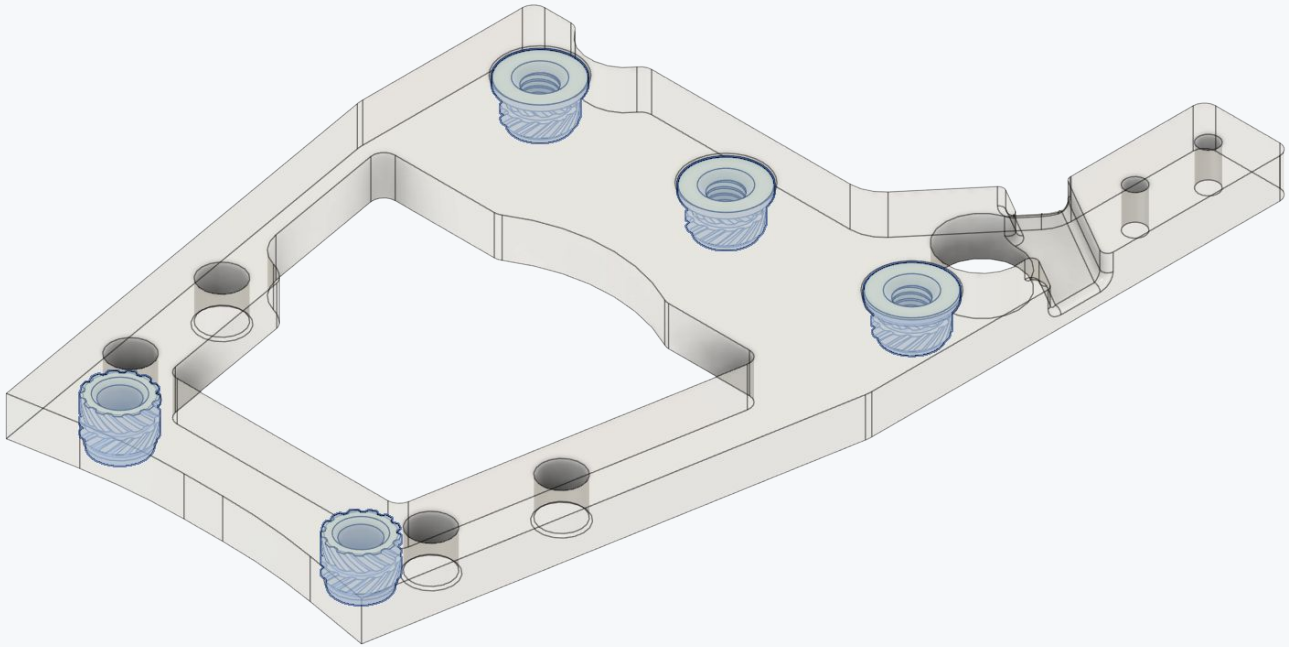


Spacer | Completed insert check

CHECK

Compare the insert positions with this transparent view.

Distinguish the flanged HYPERSETS from the standard inserts.



Spacer | Optional DF2 endstop

HARDWARE

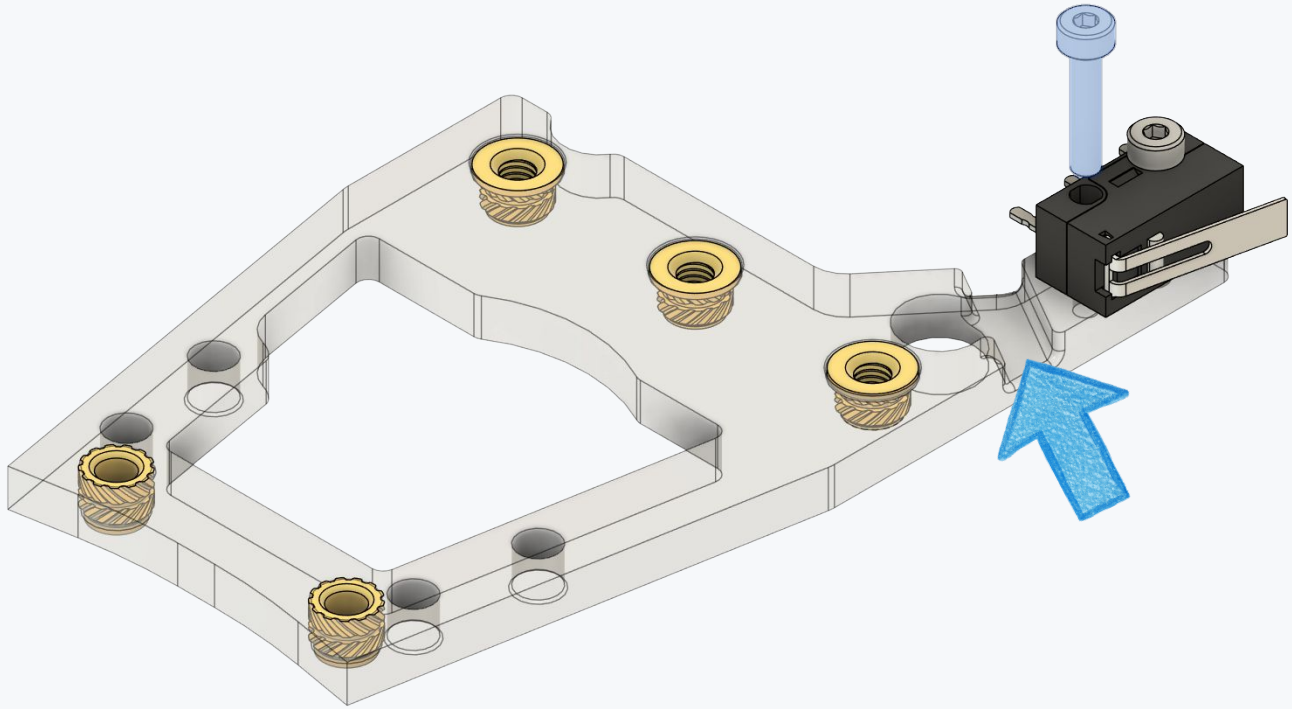
2 × M2 × 10 socket head screws

ACTION

Fit the optional endstop as shown.

CHECK

Route its wires through the indicated slot.



CAN mount brace | Inserts

HARDWARE

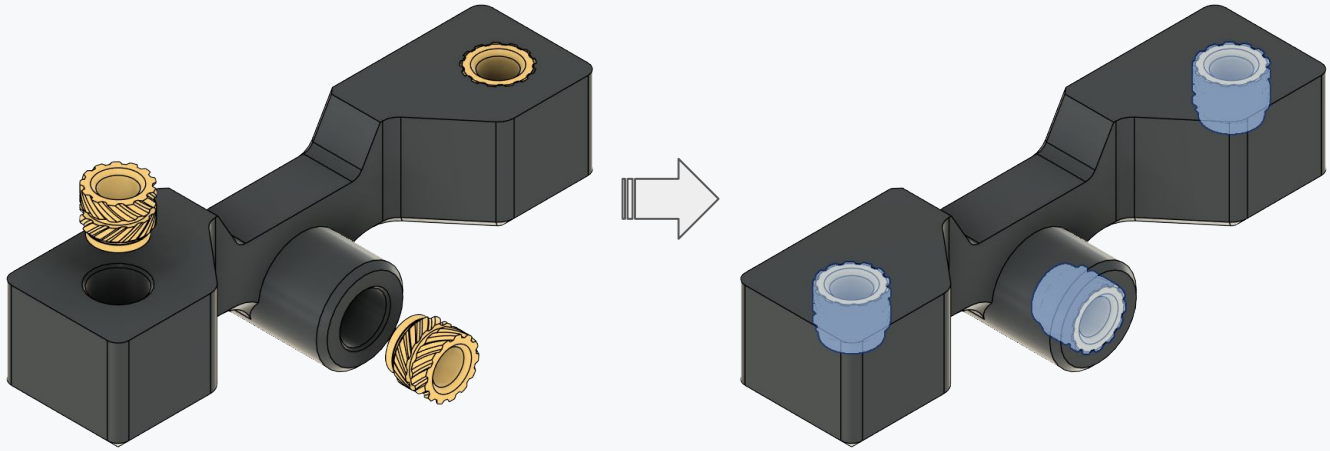
3 × M3 heat-set inserts

ACTION

Install the inserts in the illustrated holes.

CHECK

Compare the before and after views.



BEFORE AFTER

Extruder adapter | Inserts

HARDWARE

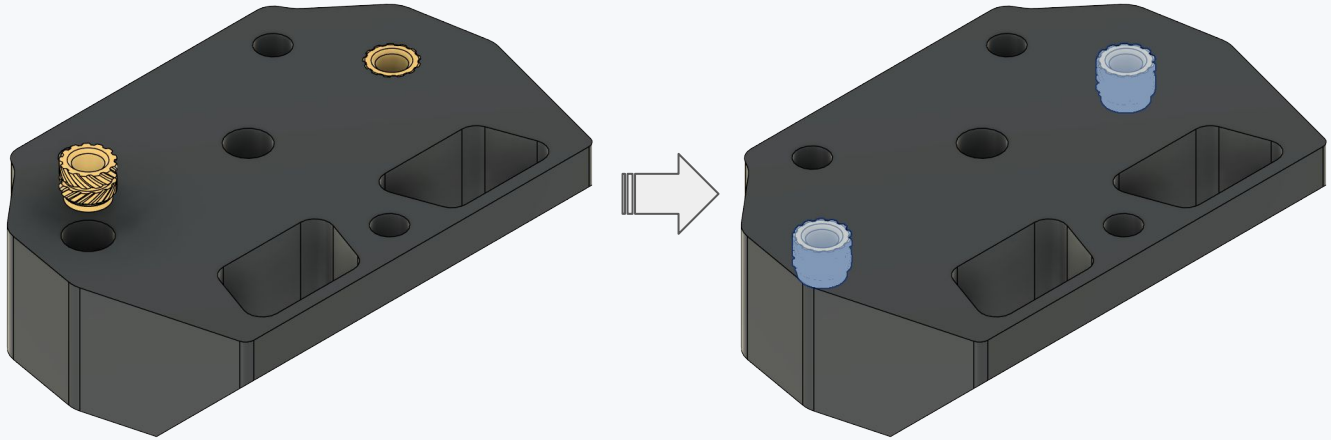
2 × M3 heat-set inserts

ACTION

Install the inserts as shown.

CHECK

Compare the before and after views.



BEFORE AFTER

CAN mount | Inserts

HARDWARE

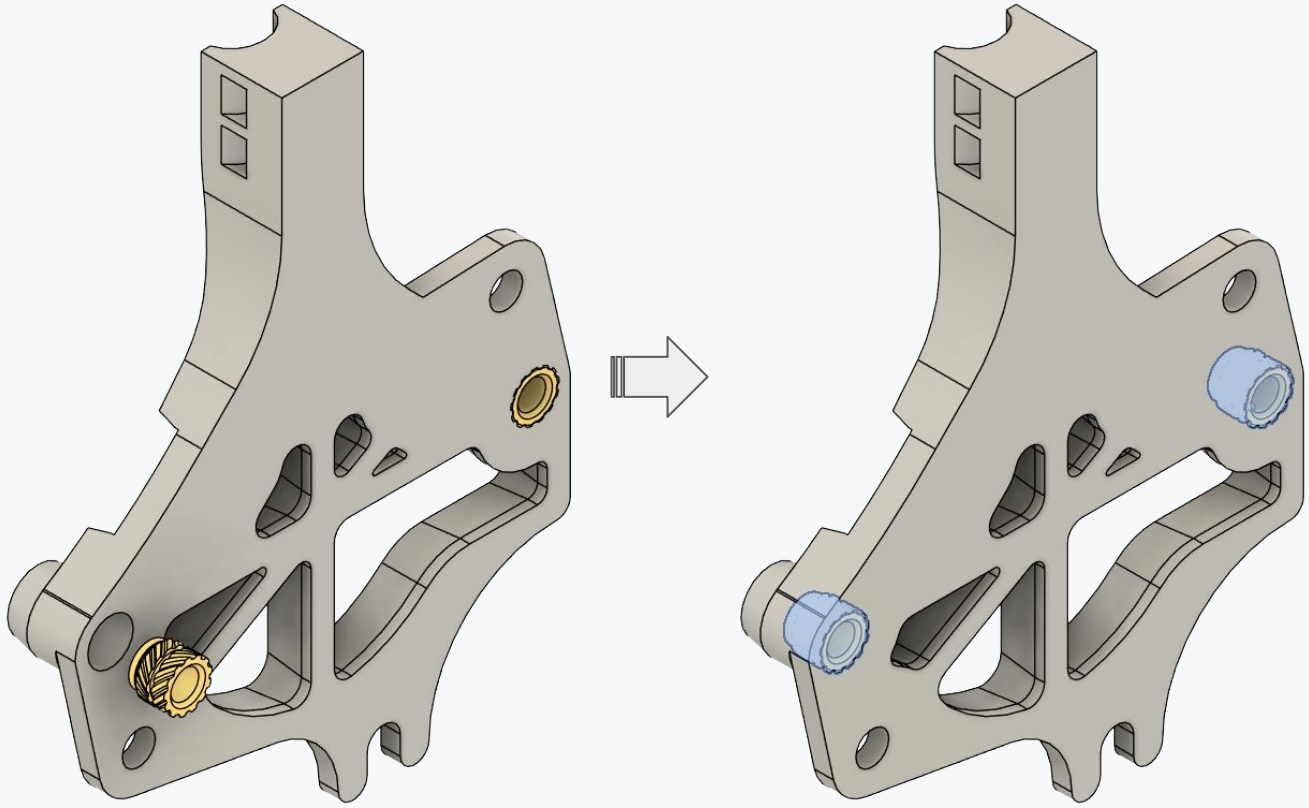
2 × M3 heat-set inserts

ACTION

Install them in the illustrated mount.

NOTE

The Sherpa Mini CAN mount does not need these inserts.



BEFORE AFTER

Mount the NovaCore

HARDWARE

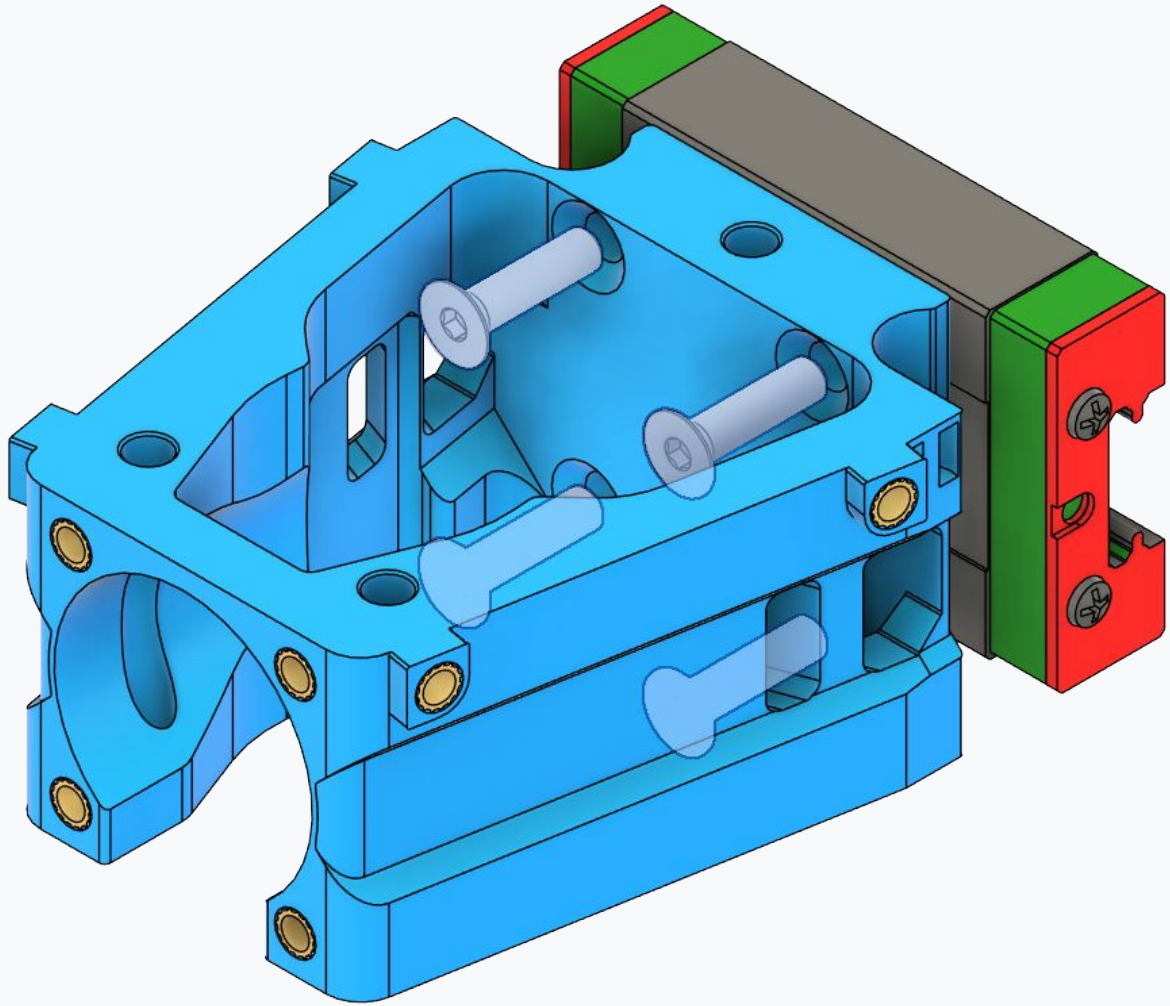
4 × M3 × 12 screws

ACTION

Mount the core as shown.

CHECK

Align the mounting holes before tightening.



Attach the CAN mount brace

HARDWARE

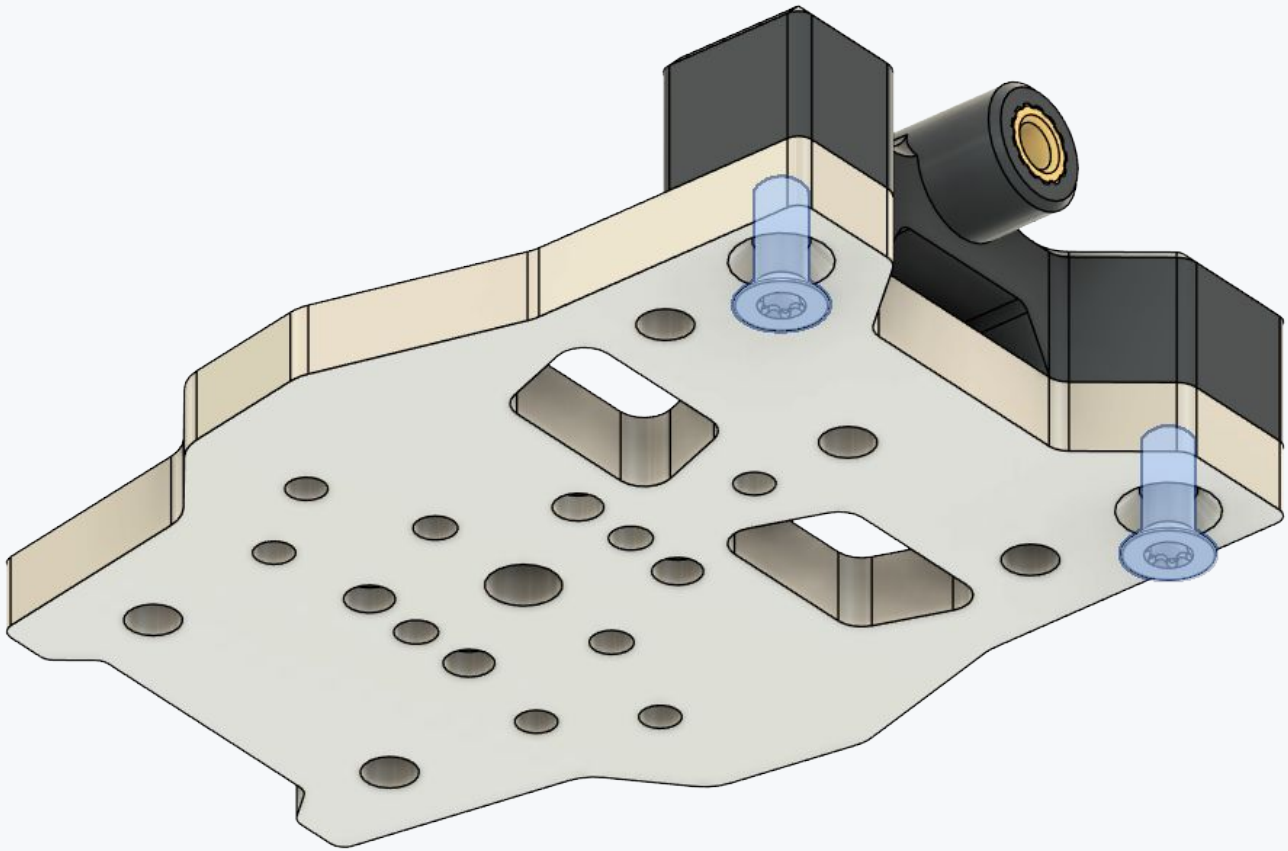
2 × M3 × 8 countersunk screws

ACTION

Attach the brace to the top plate.

CHECK

Match the orientation shown.



Fit the top plate to the core

HARDWARE

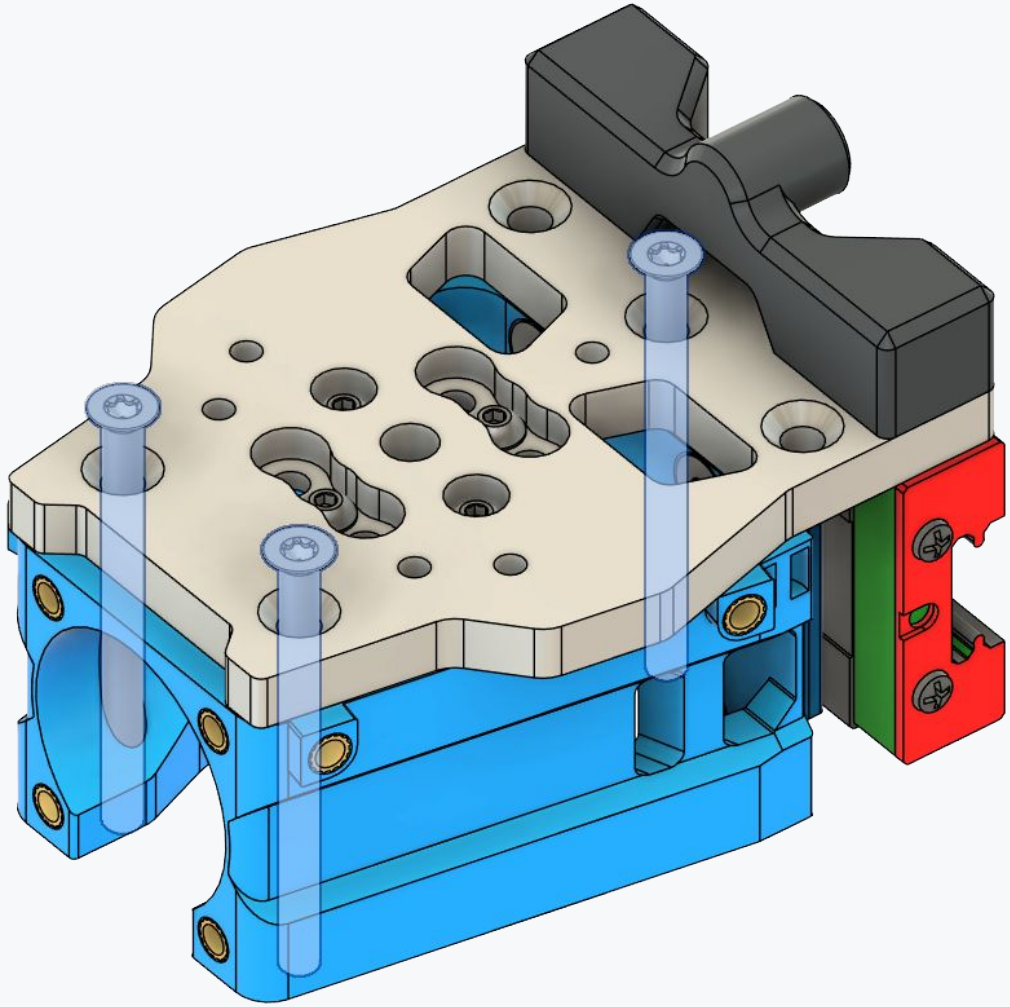
3 of the 5 M3 × 40 countersunk screws

ACTION

Insert the screws through the top plate.

CHECK

Use the three illustrated holes.



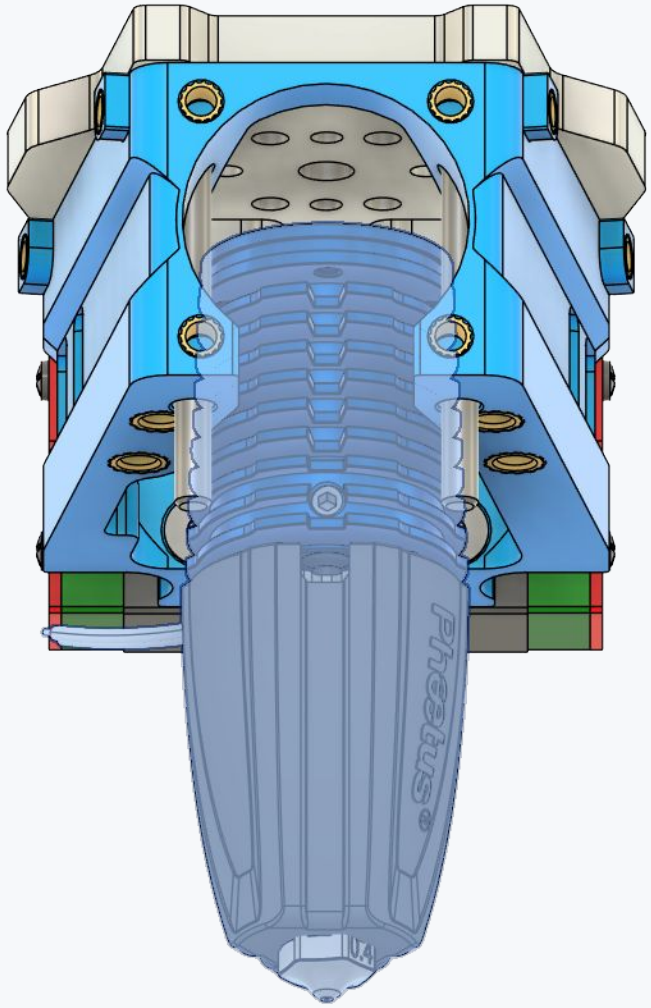
Position the hotend

ACTION

Align the hotend with the core. Route the wires to the left in this view.

NOTE

The heater block may need rotation. Follow the hotend maker’s procedure.



Secure the hotend

HARDWARE

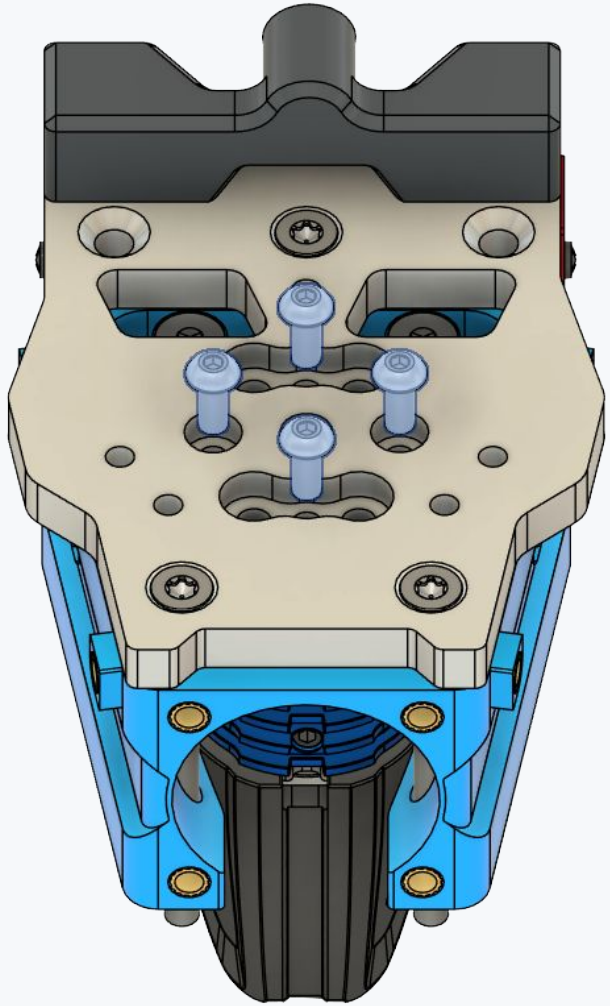
4 × M2.5 × 6 screws

ACTION

Mount the hotend as shown.

CHECK

Keep the wires clear of mating surfaces.



Fit the Beacon standoffs

HARDWARE

2 × M3 × 8 countersunk Torx screws

STANDOFF LENGTH

26 mm: UHF-length hotends

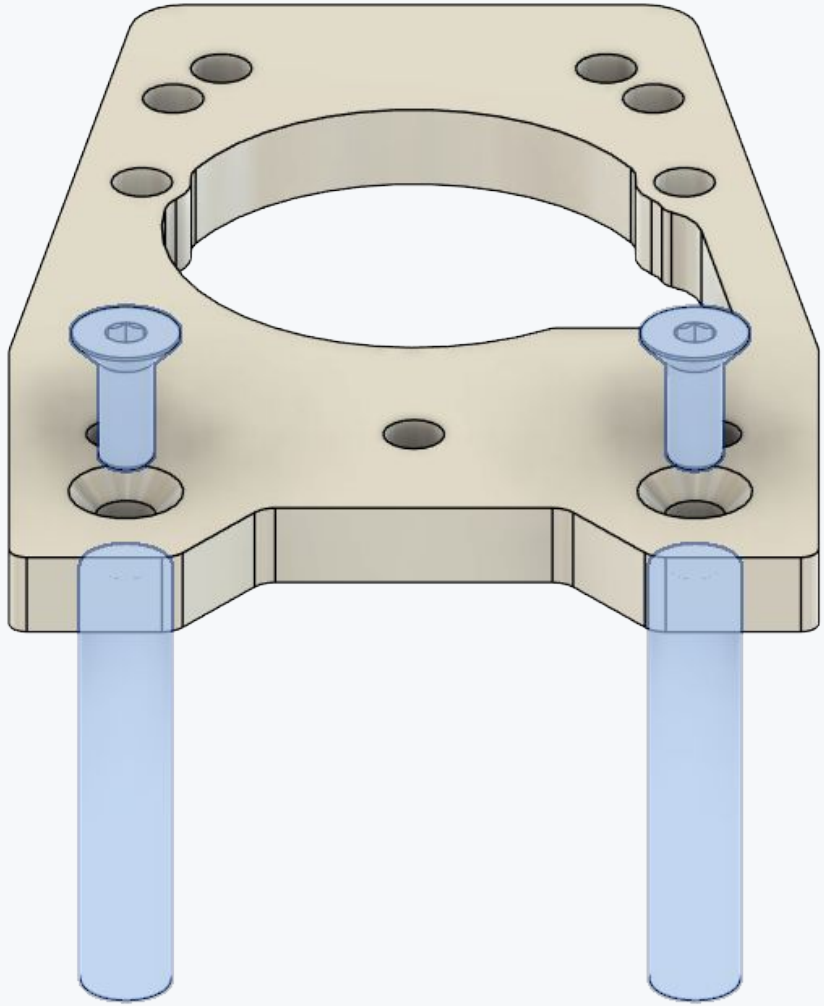
30 mm: Chube

ACTION

Install the screws into the Beacon standoffs.

CHECK

Match the bottom-plate orientation shown.



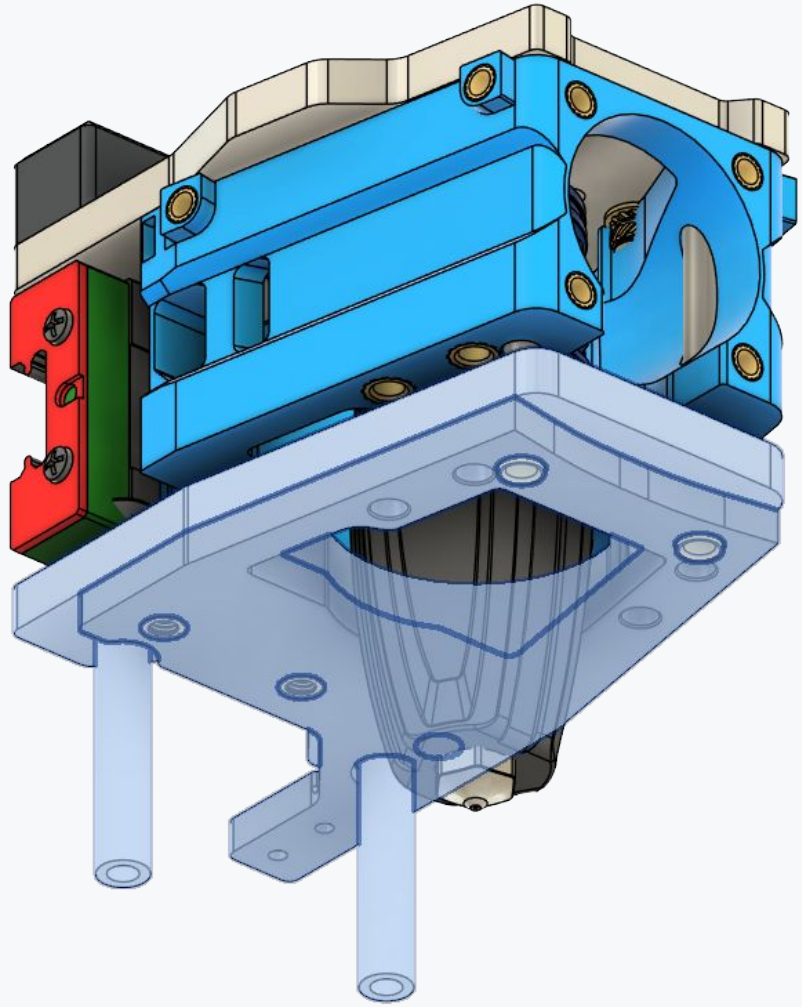
Position the bottom plate

ACTION

Align the bottom plate and spacer with the core.

CHECK

Align the screw paths and keep the hotend wires clear.



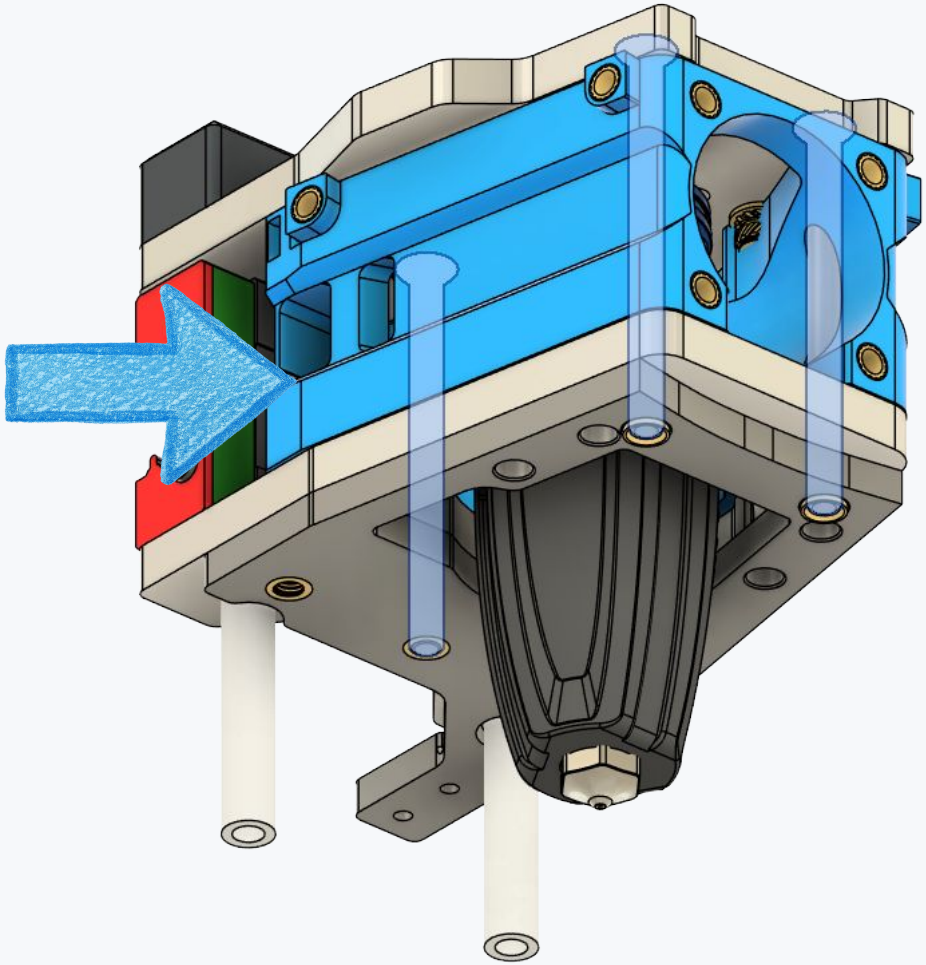
Install the three top screws

ACTION

Install the three top screws shown.

CHECK

Route the hotend wires through the indicated exit. Keep them clear of the plate and screw paths.



Install the 2510 fan and cover

HARDWARE

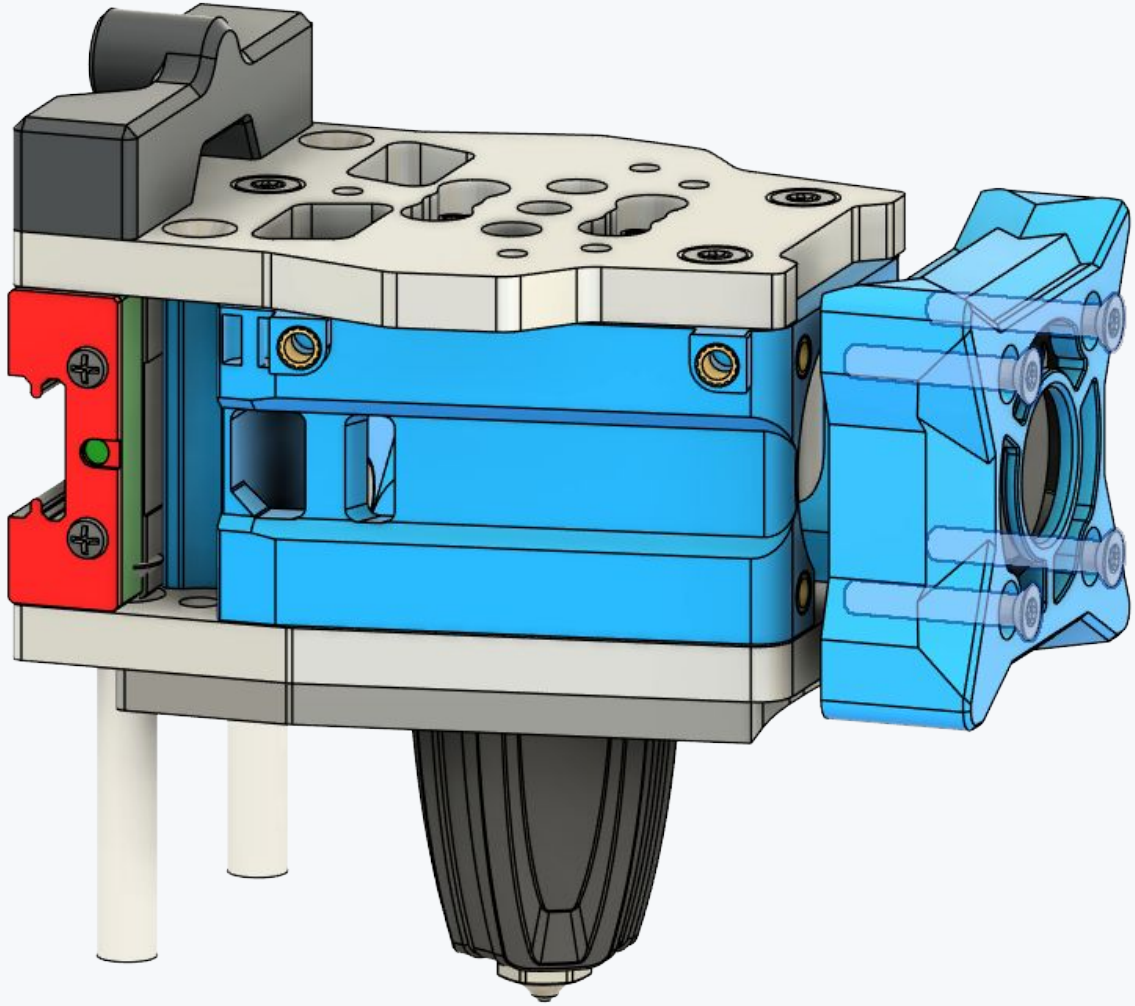
4 × M2.5 × 16 countersunk Torx screws

ACTION

Seat the 2510 fan in the cover, then install the screws.

CHECK

Keep the fan wires clear.



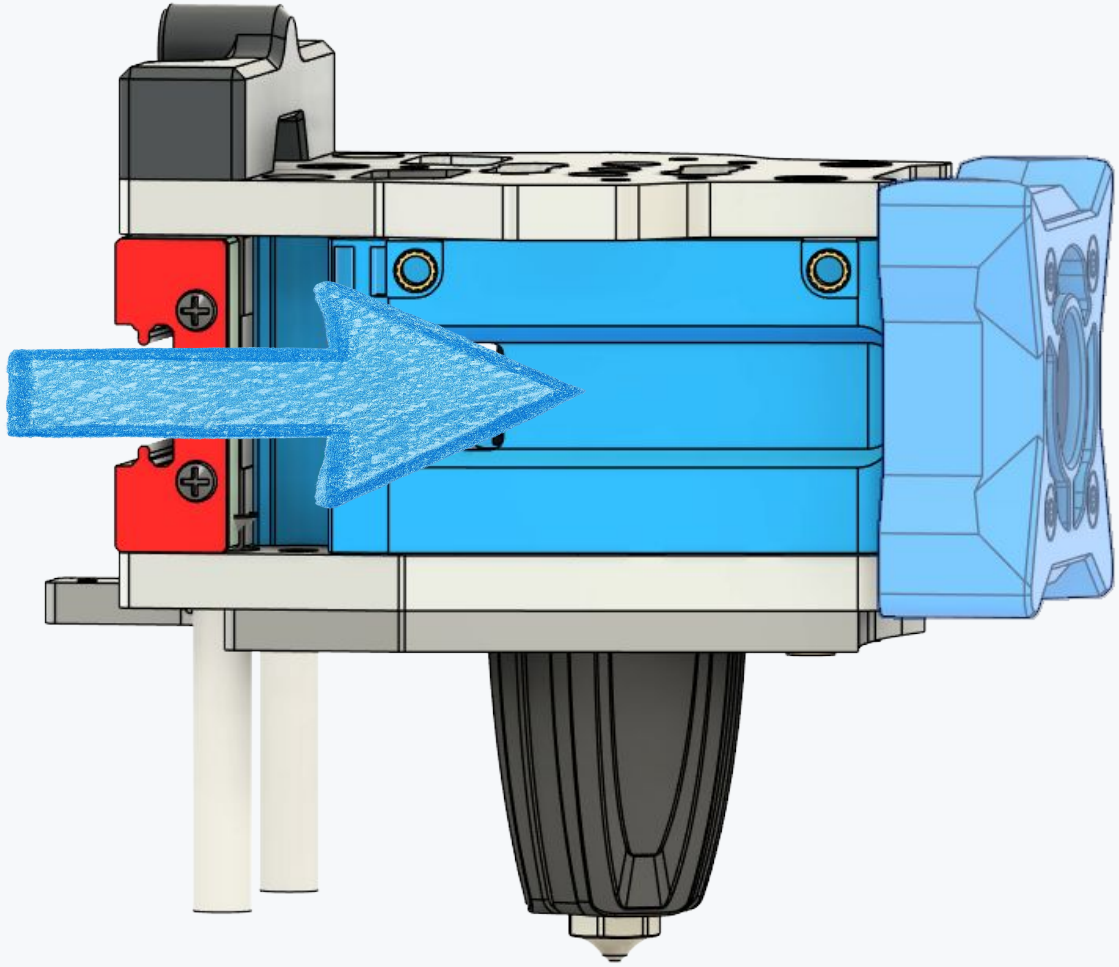
Route the fan wires

ACTION

Use the indicated channel for the fan wires.

CHECK

Keep the wires clear of the fan blades and mating surfaces.



Mount the fan ducts

HARDWARE

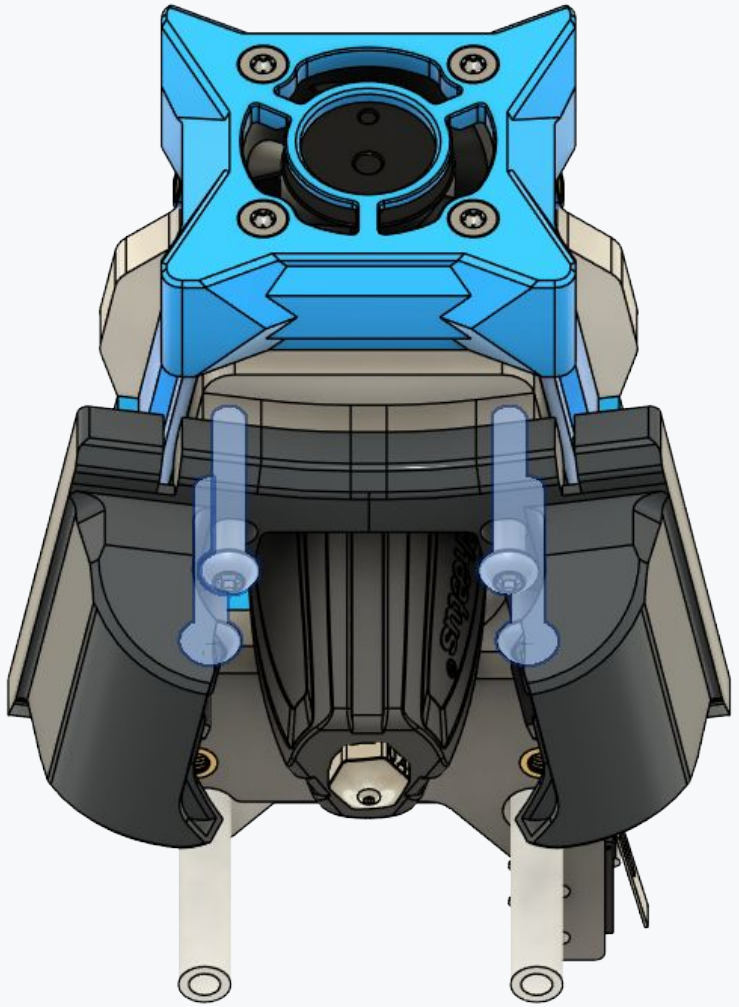
4 × M3 × 20 button head Torx screws

ACTION

Mount the ducts as shown.

NOTE

The next step may require loosening these screws to seat the fans.



Seat the 4010 fans

ACTION

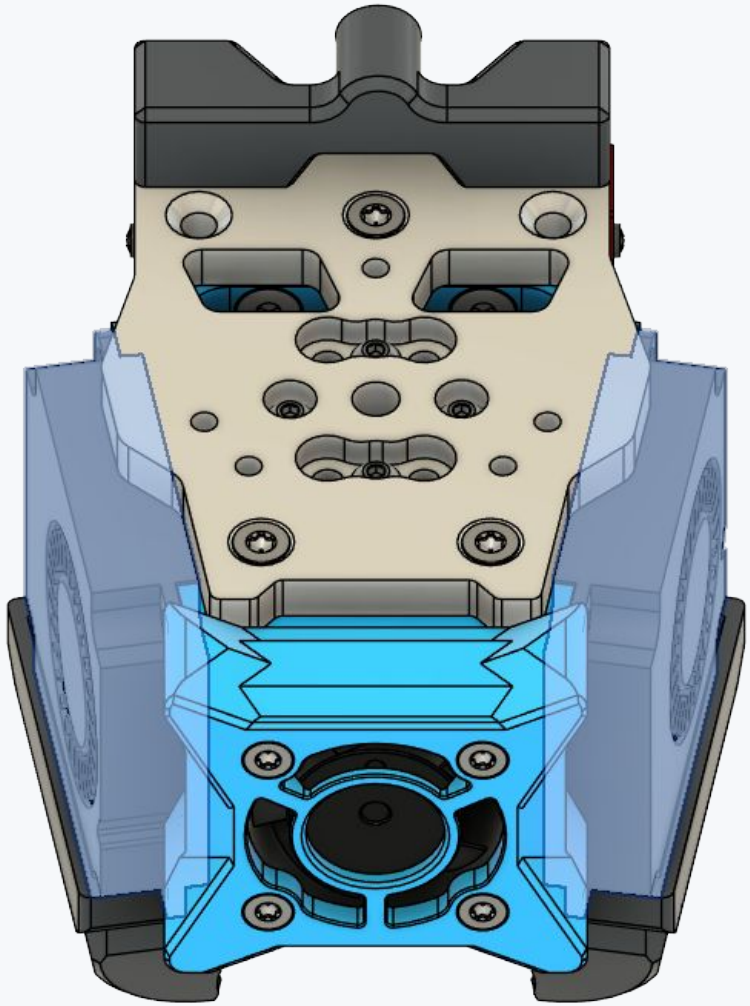
Angle the fans into the ducts, then press them into the core.

NOTE

Loosen the duct screws if needed for fit.

CHECK

Seat the fans without trapping wires.



Install the top fan hardware

HARDWARE

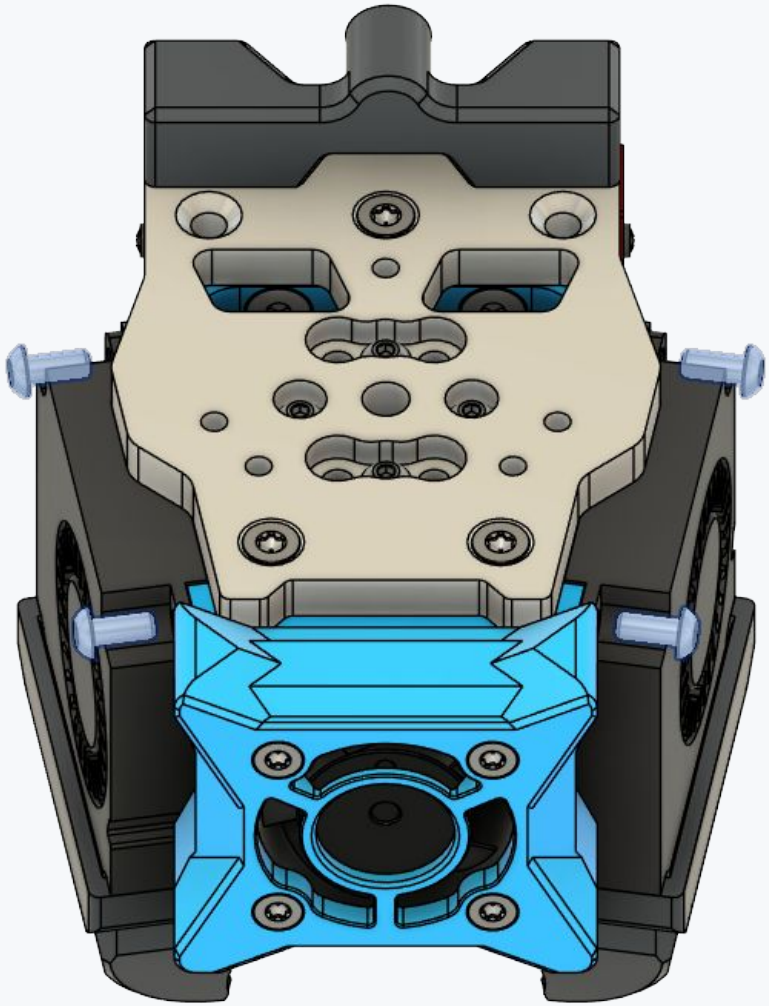
4 × M2.5 × 6 button head Torx screws

ACTION

Install the screws in the positions shown.

CHECK

Keep the fan housings aligned.



Tighten the top fan hardware

HARDWARE

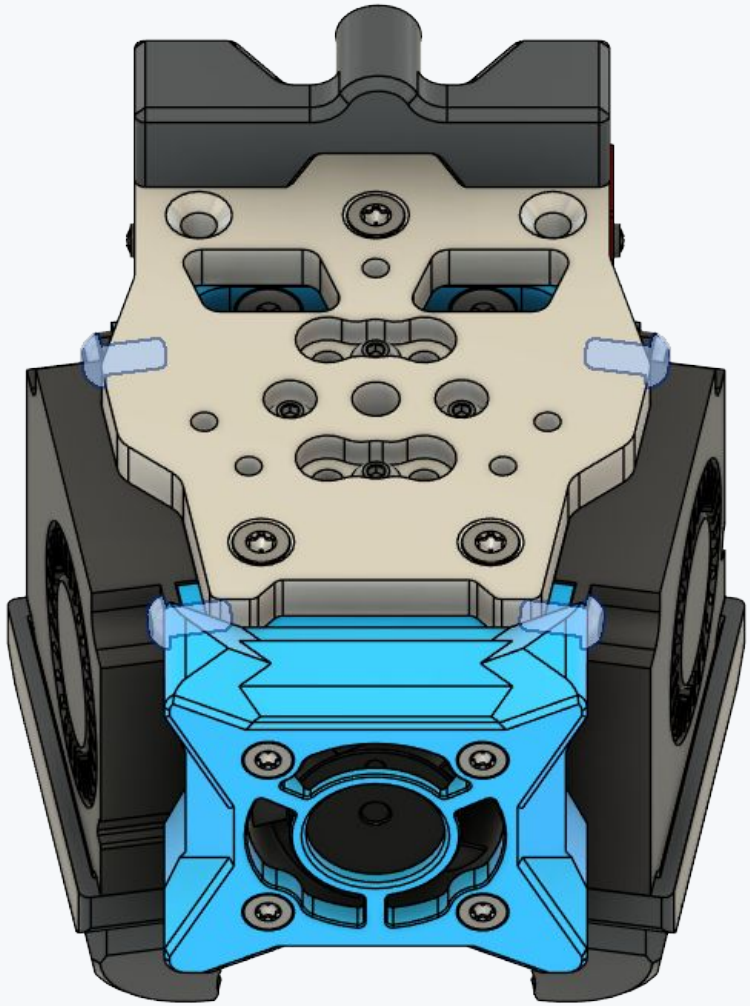
4 × M2.5 × 6 button head Torx screws

ACTION

Finger-tighten the screws.

NOTE

Do not overtighten. No torque value is specified.



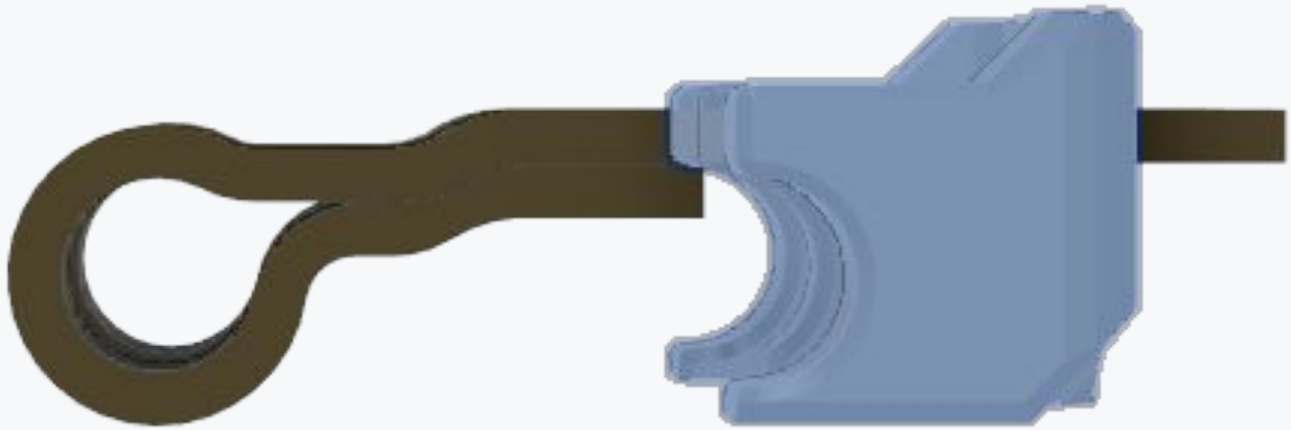
Belts | Fold the belt end

ACTION

Insert the belt end into the belt tie. Fold the belt back with the teeth facing each other.

CHECK

Match the loop orientation shown.



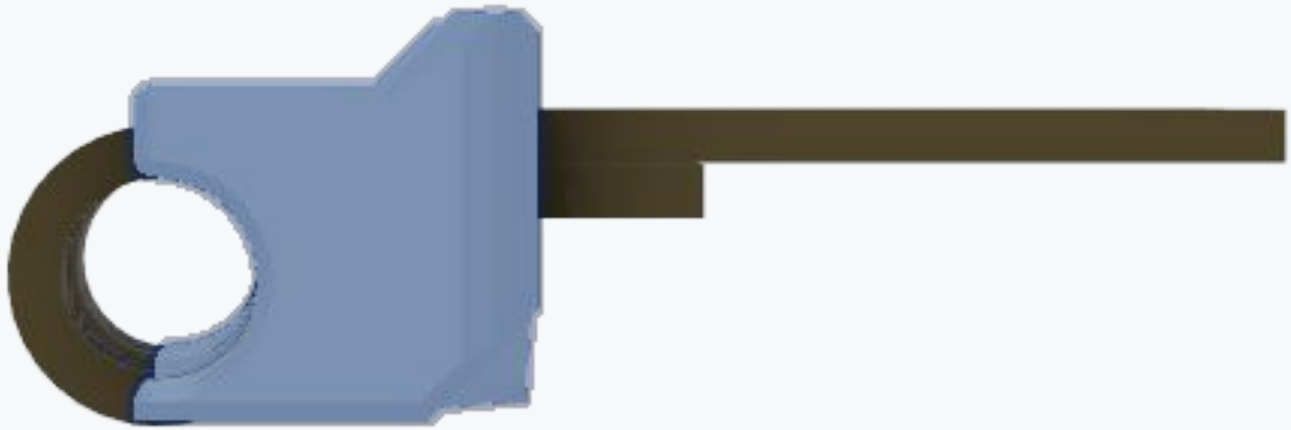
Belts | Position the belt tie

ACTION

Slide the belt tie over the folded belt loop.

CHECK

Keep the folded sections aligned.



Belts | Insert the spacer

HARDWARE

25 mm belt-tie spacer

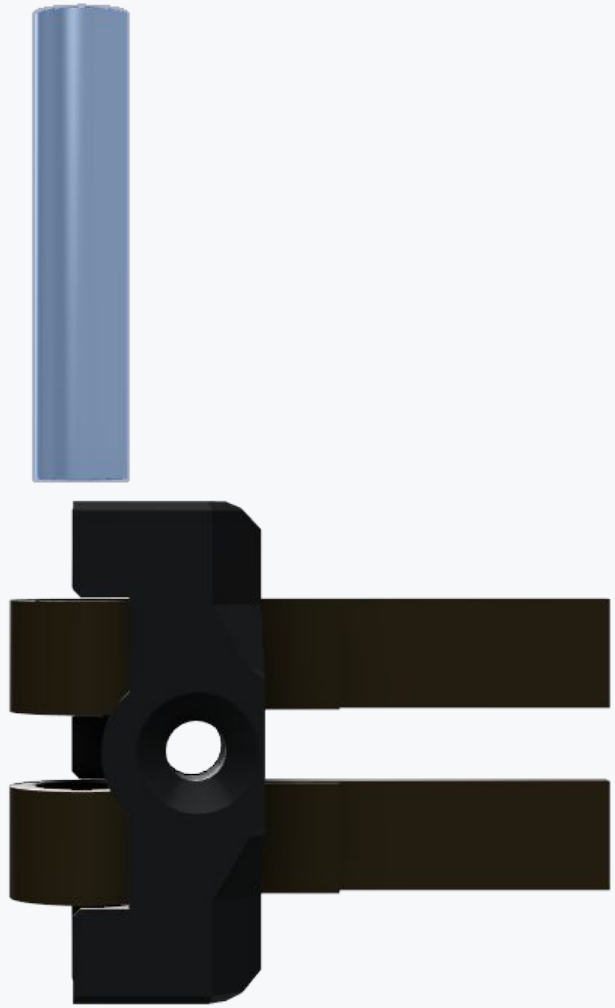
2 supplied in the kit

ACTION

Slide the spacer through the belt loop.

CHECK

Center it in the loop. This is a different part from the Beacon probe standoffs.



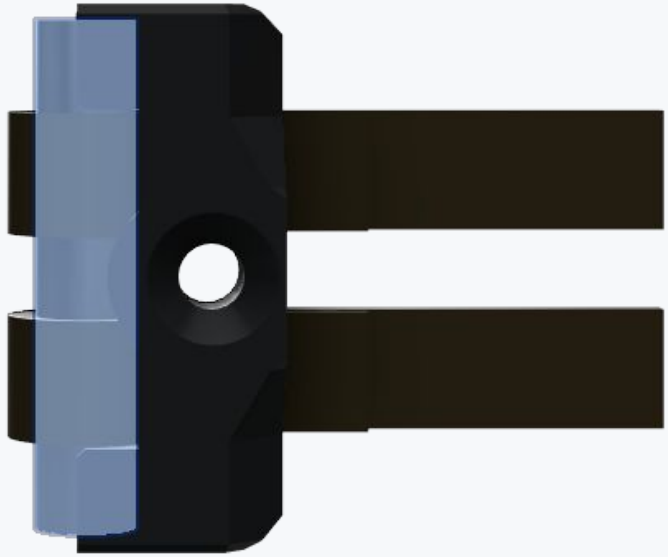
Belts | Center the spacer

ACTION

Center the spacer, then pull the belts tight around it.

CHECK

Keep the folded belt sections aligned.



Belts | Insert the clamp screw

HARDWARE

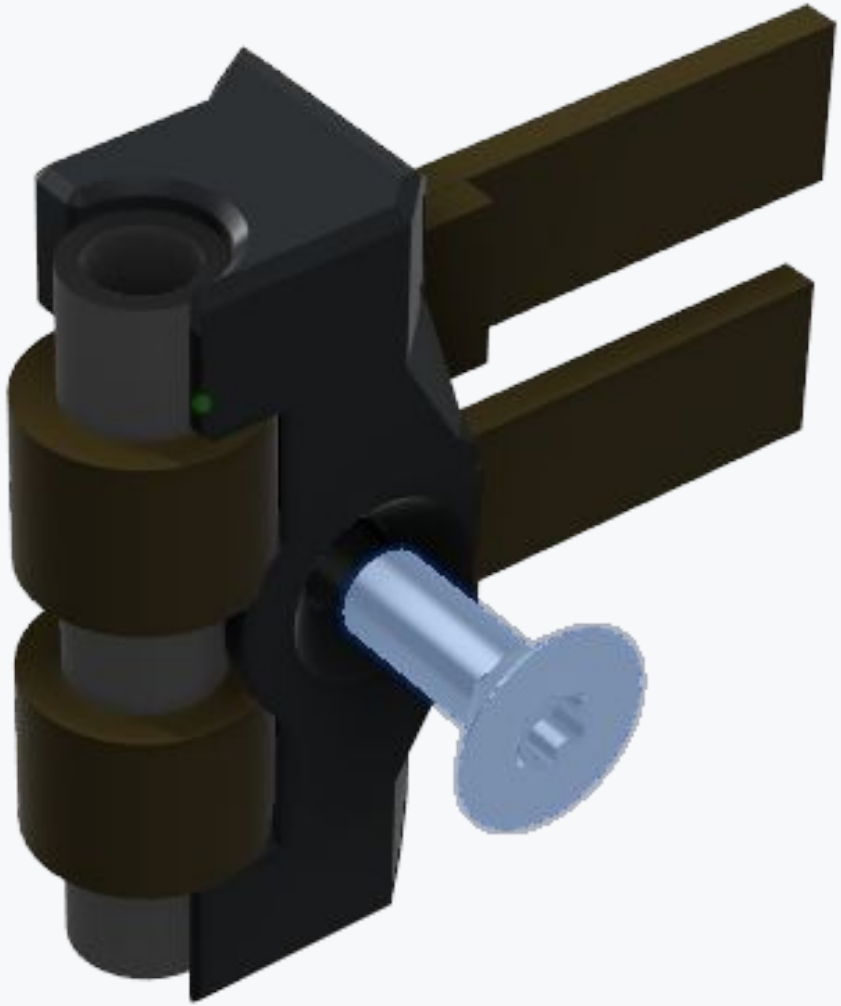
M3 × 8 countersunk Torx screw

ACTION

Insert the screw from the side shown.

CHECK

Keep the belt tie and spacer aligned.

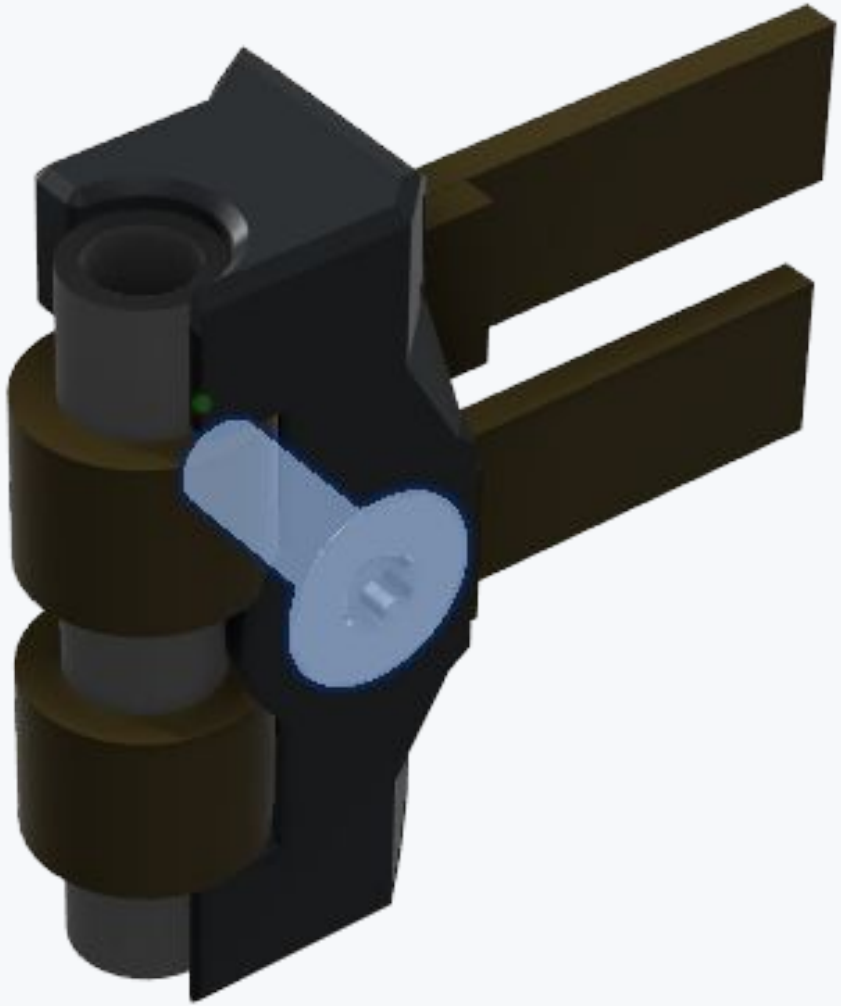


Belts | Check screw position

CHECK

Compare the screw position with this view before fitting the T-nut.

Keep the belt loop seated around the spacer.



Belts | Fit the MakerBeam XL T-nut

HARDWARE

MakerBeam XL T-nut

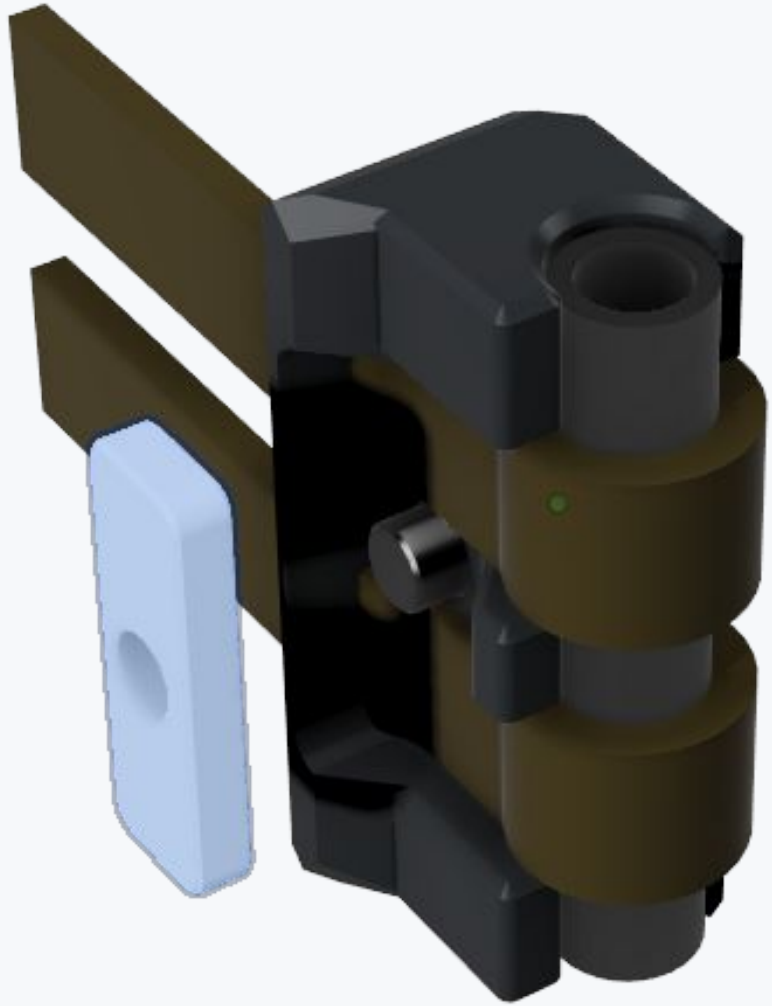
2 supplied in the kit

ACTION

Fit the T-nut in the orientation shown.

CHECK

Keep the screw and T-nut aligned.

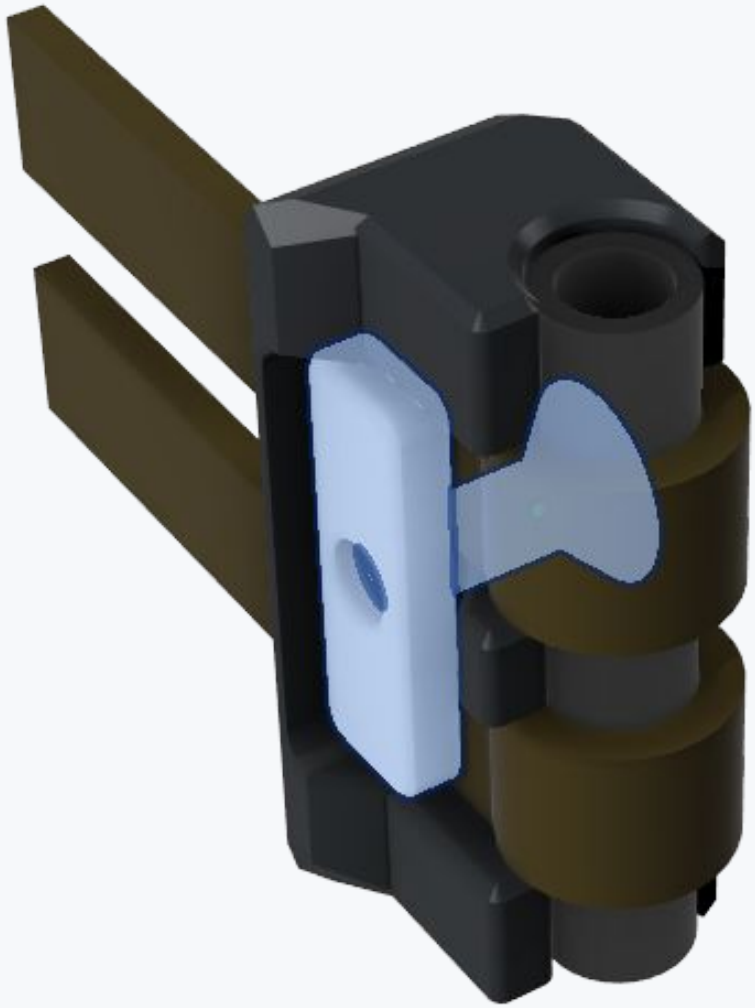


Belts | Check the T-nut

CHECK

Compare the T-nut position with this completed view.

Keep the belt tie, screw and T-nut aligned.

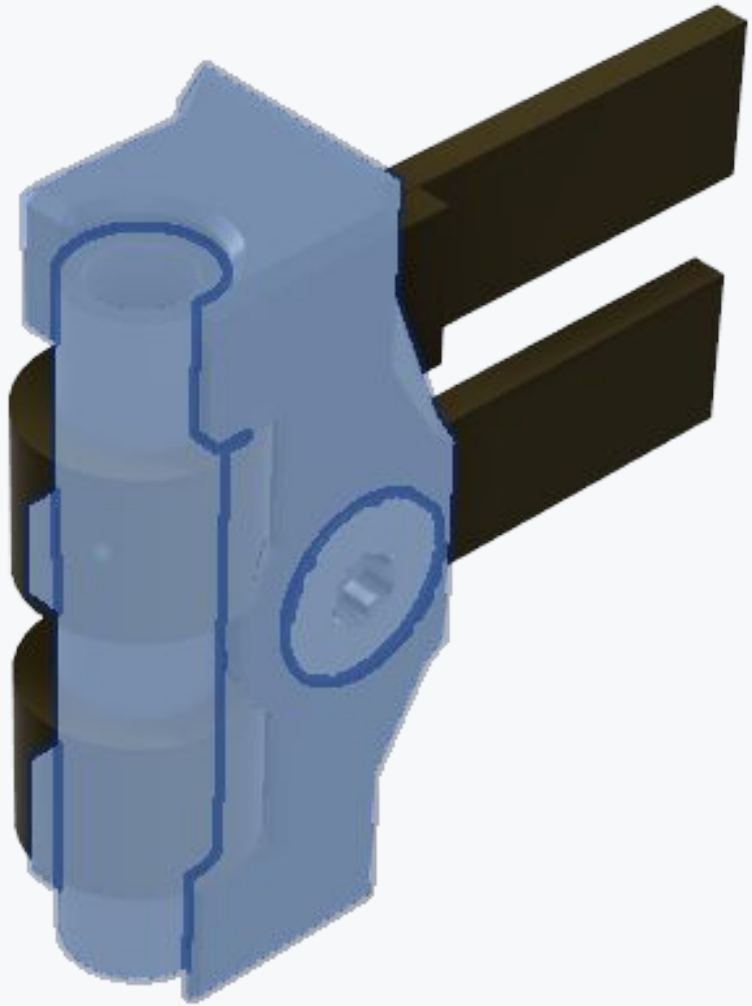


Belts | Completed clamp check

CHECK

Compare the assembled clamp with this transparent view.

Confirm the belt loop, spacer and hardware are in their illustrated positions.



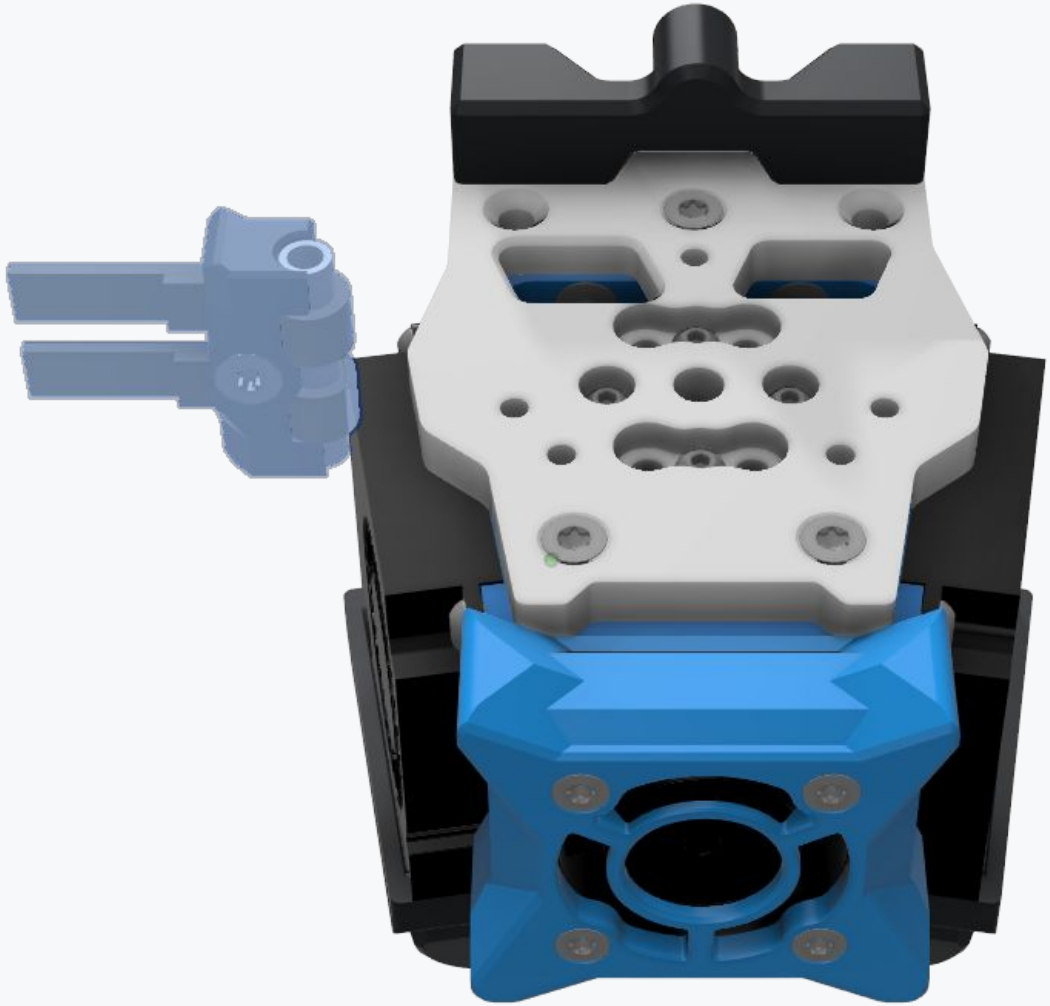
Belts | Position the first clamp

ACTION

Position the first belt clamp on the toolhead as shown.

CHECK

Align the clamp with the top screw path.



Belts | Install the first top screw

HARDWARE

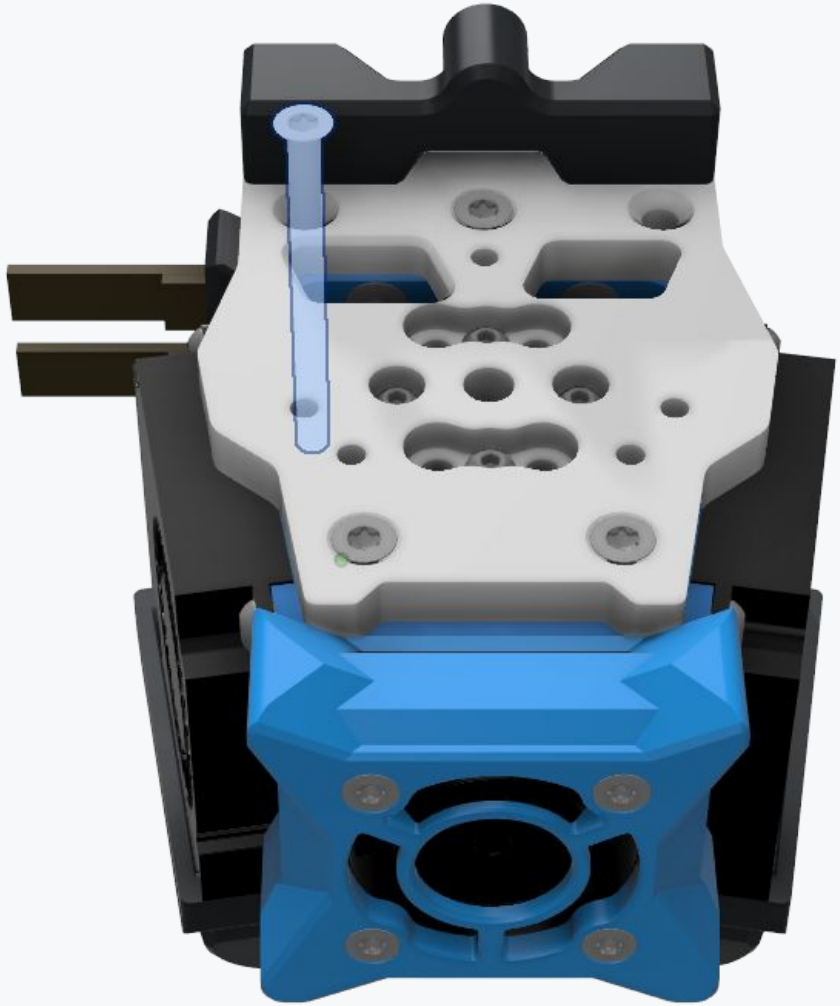
1 × M3 × 40 countersunk Torx screw

ACTION

Hold the belt tie in place and insert the screw.

CHECK

Keep the clamp aligned.



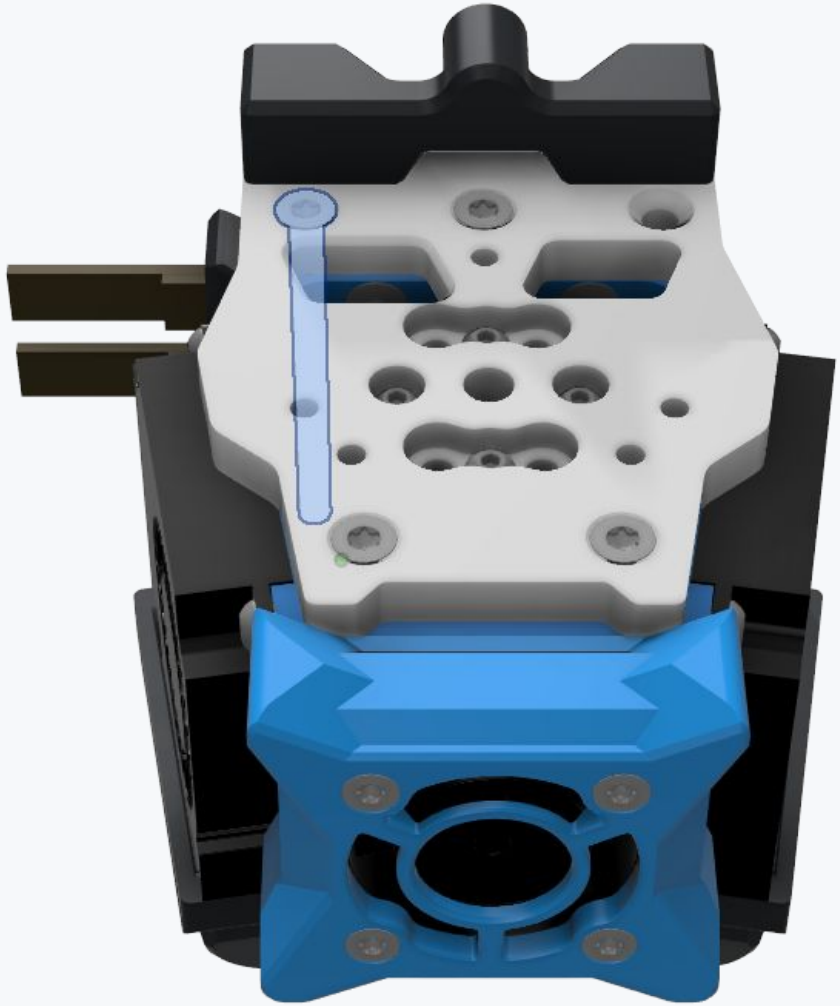
Belts | Secure the first clamp

ACTION

Hold the belt tie in place while tightening the M3 × 40 countersunk Torx screw.

CHECK

Keep the clamp seated in the illustrated position.



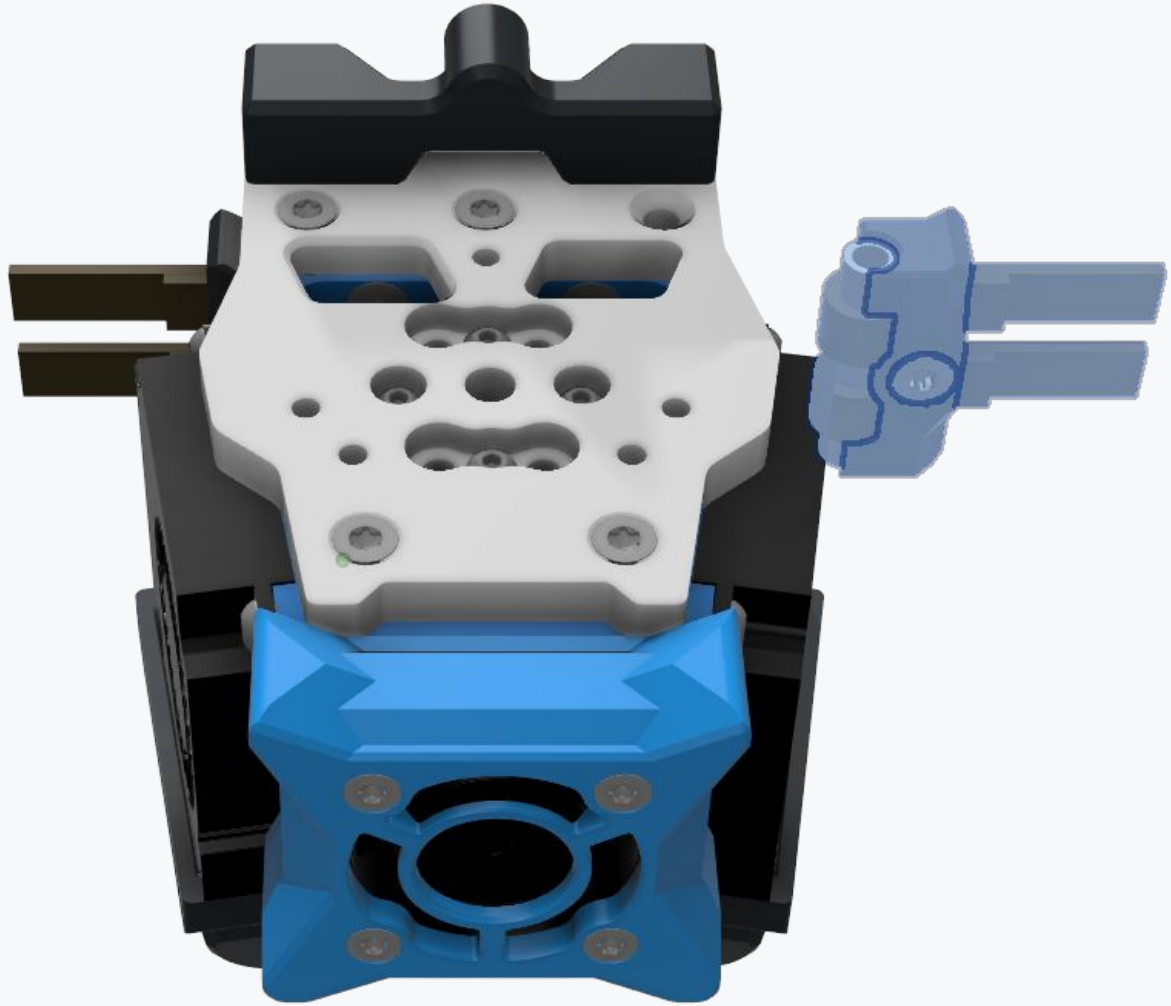
Belts | Position the opposite clamp

ACTION

Position the opposite belt clamp as shown.

CHECK

Align it with the remaining top screw path.



Belts | Install the opposite top screw

HARDWARE

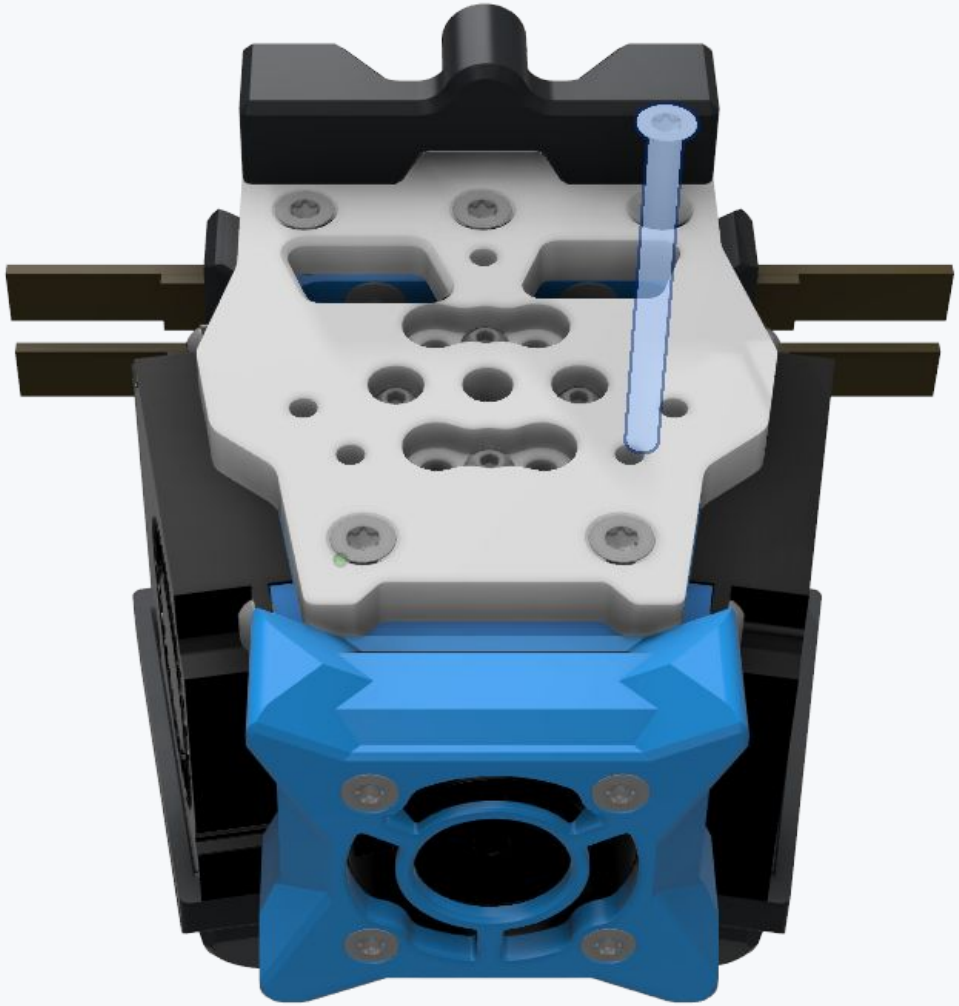
1 × M3 × 40 countersunk Torx screw

ACTION

Hold the belt tie in place and insert the screw.

CHECK

Keep the clamp aligned.



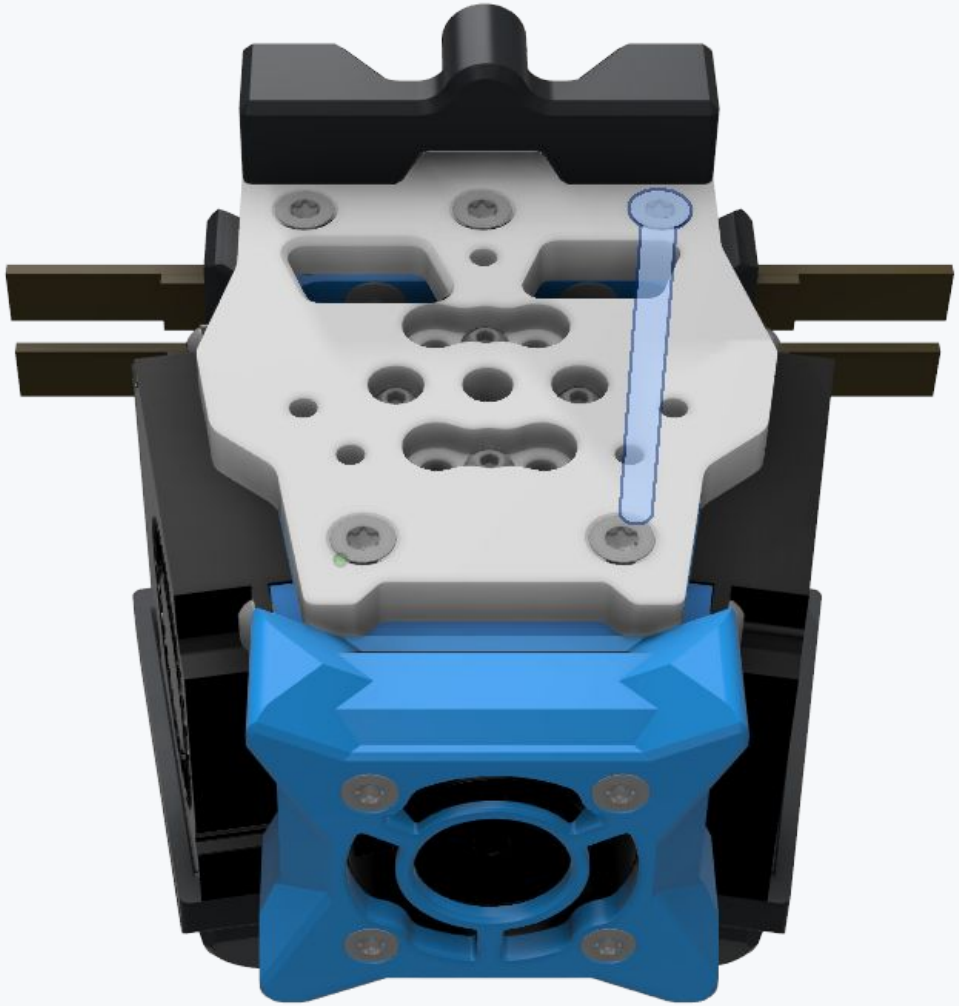
Belts | Secure the opposite clamp

ACTION

Hold the belt tie in place while tightening the M3 × 40 countersunk Torx screw.

CHECK

Keep the clamp seated in the illustrated position.



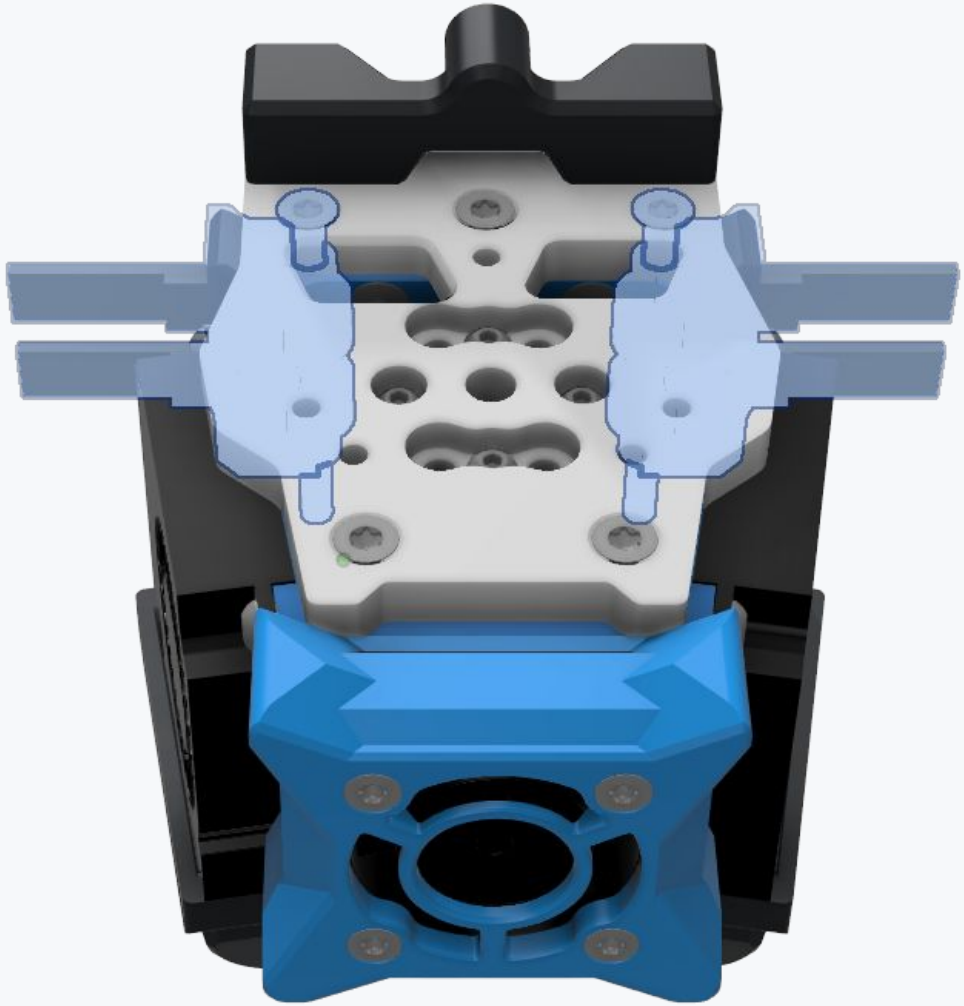
Belts installed | Inspect both sides

CHECK

Compare both belt clamps with the completed view.

Check that the belts are untwisted and the clamps are seated.

Continue to the final mechanical inspection.



Final mechanical inspection

BEFORE YOU CONTINUE

Compare the assembly with the completed views.

Check that the mounted parts and belt clamps are seated.

Check for trapped wires at plates, ducts and fan housings.

Account for unused hardware and keep spare parts together.

NOT INCLUDED IN THIS GUIDE

Electrical connections and board-specific pinouts

Firmware configuration and sensor setup

Belt tensioning and printer calibration

Complete those procedures using the documentation for your actual components before operation.

Mechanical assembly complete does not mean ready to power on or print.

REFERENCES

Klippain ShakeTune

Vibration-analysis tools

Ellis' Print Tuning Guide

Printer tuning reference

Xol Toolhead

Design inspiration and project reference

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